



Computer Vision for Autonomous Vehicles

Beyond The camera

Instructor: Golnaz Habibi, PhD

University of Oklahoma

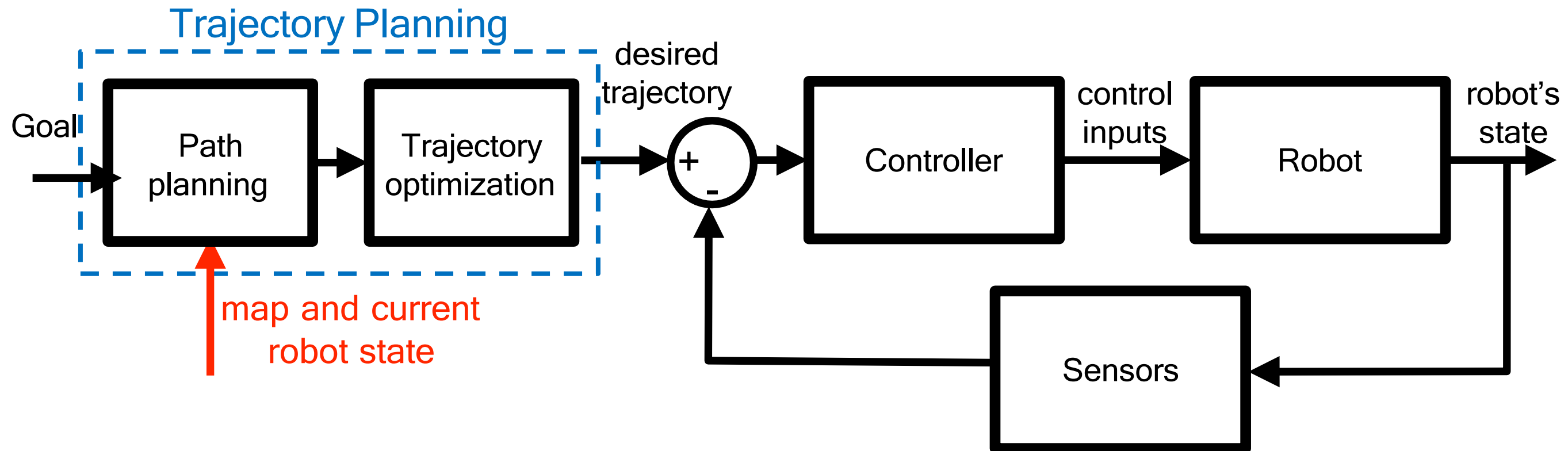
Fall 2025



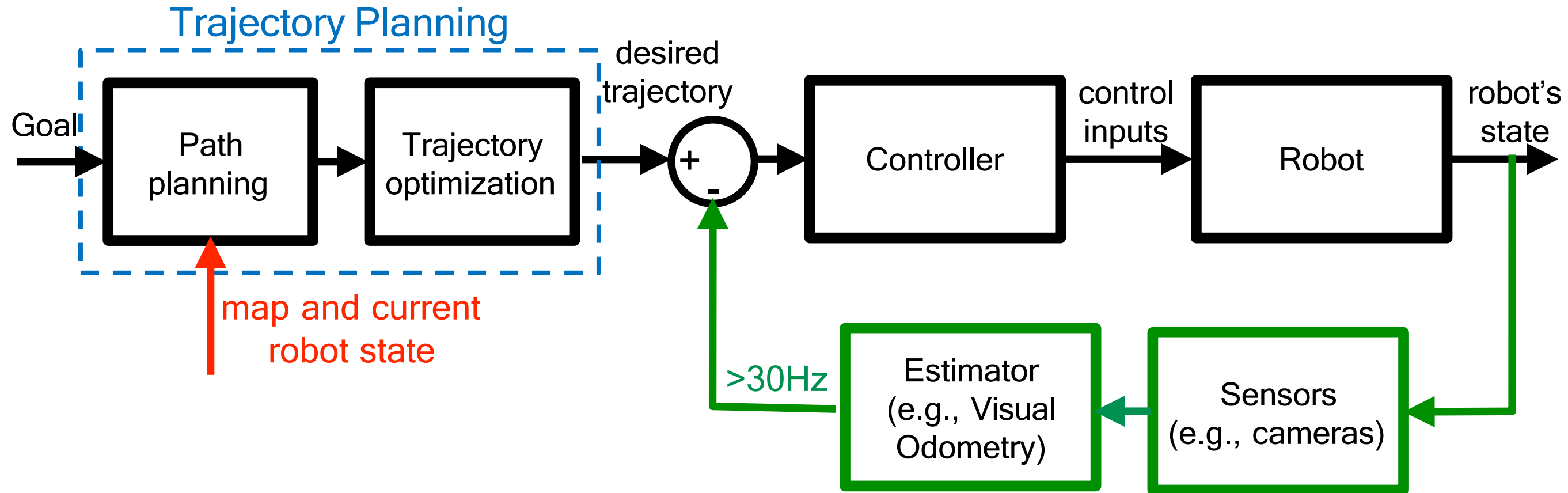
Announcement

- Lab 6 grade will be posted today
- Last Quiz solution will be posted today
- Poster presentation this Friday (12- 1): you need to send an email by tomorrow Dec 10, 5 PM
- Final project presentation next Tuesday on December 16th, ALL member need to present
- Class Reflection survey by Sunday
- 80% or more → everyone will get +3 in final grade
- Today: summary of the class and 3D construction and RL

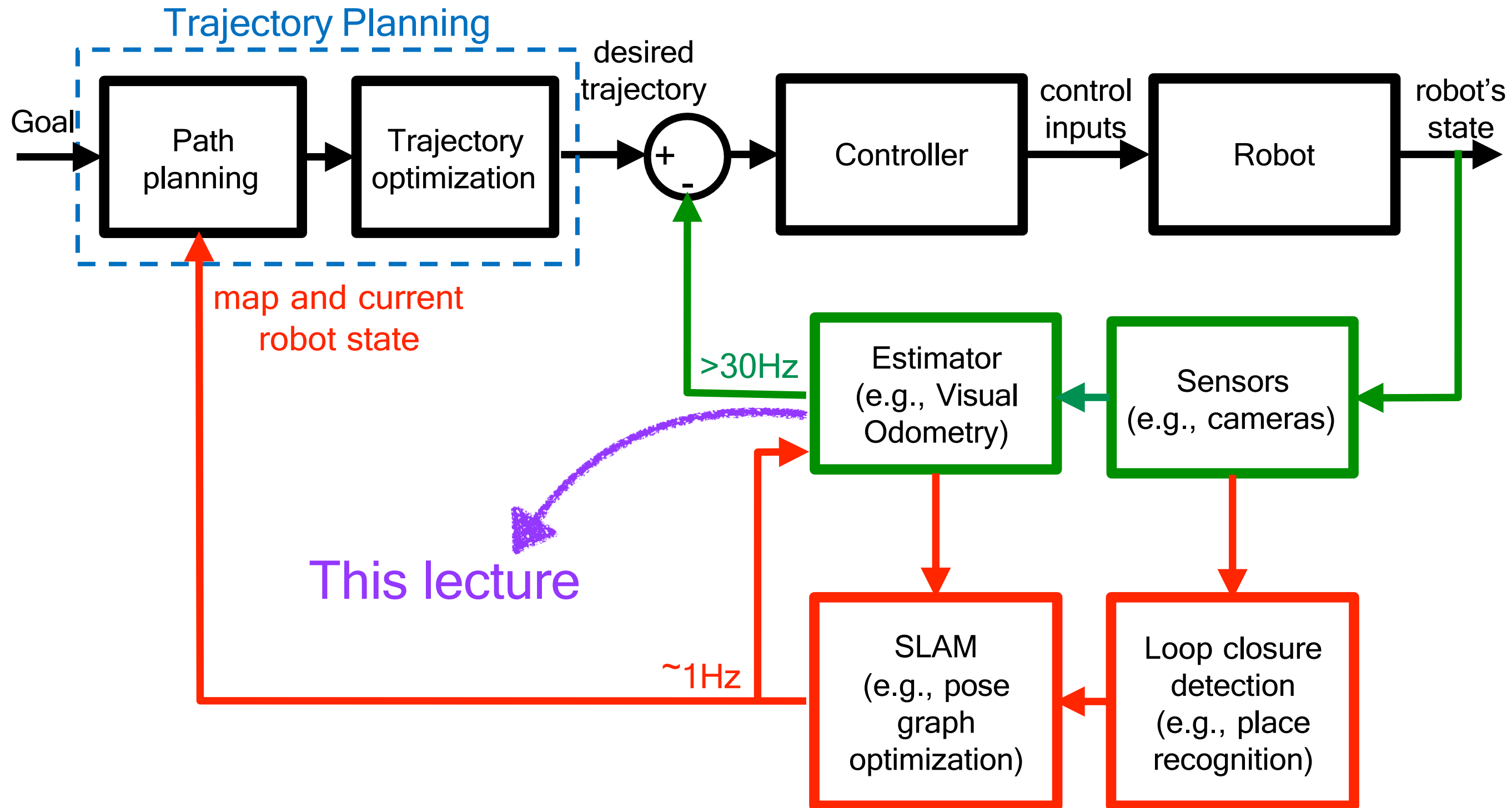
Big Picture



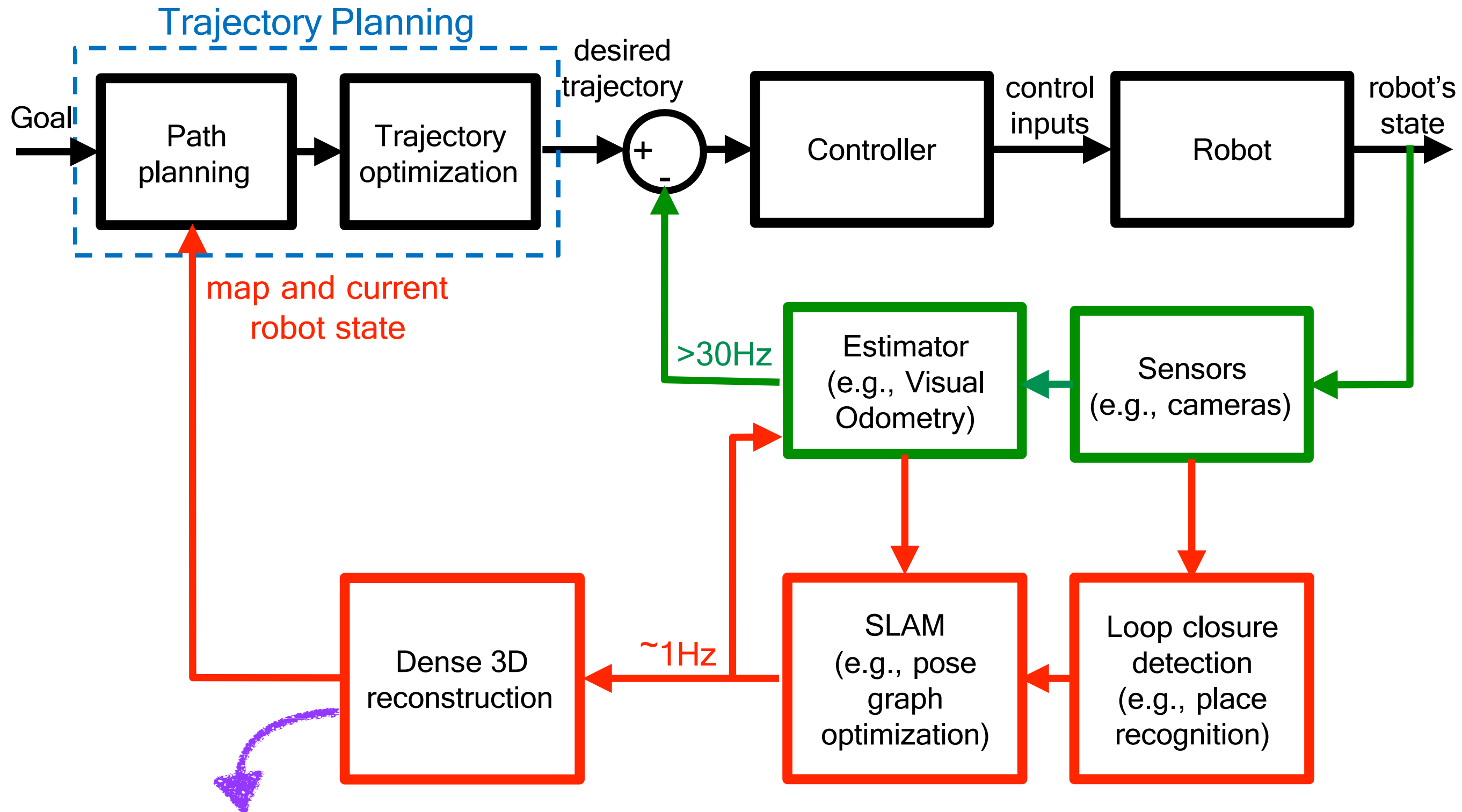
Big Picture



Big Picture



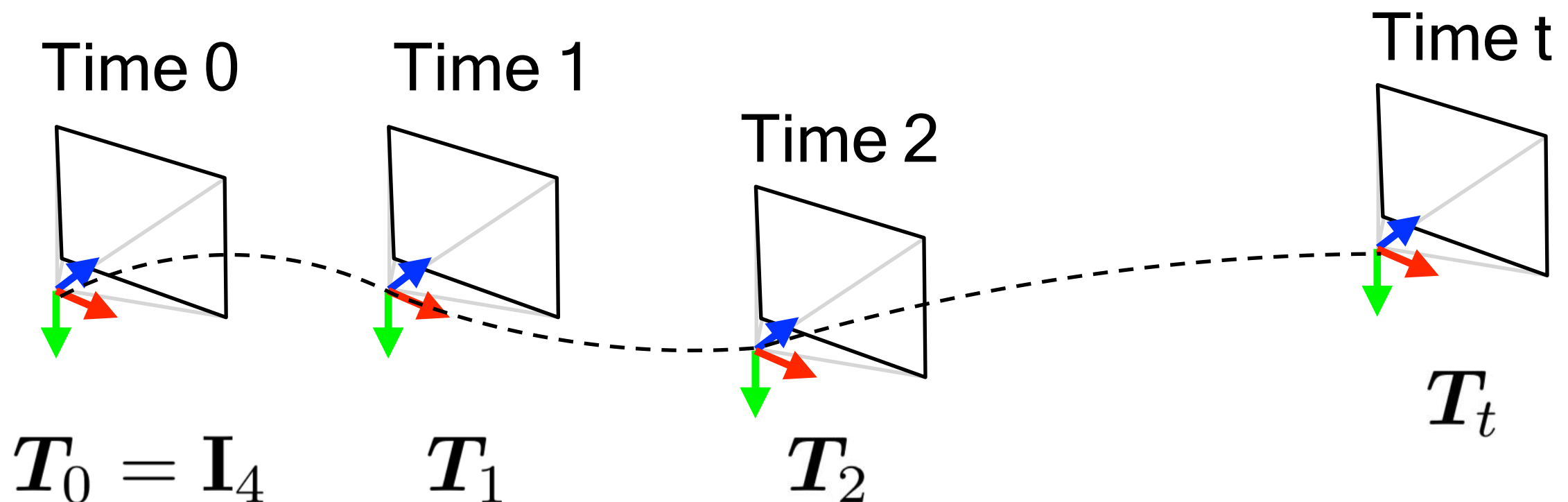
Big Picture



Next lecture

Previously on VNAV: Visual Odometry

Visual odometry (VO): motion estimation estimation based on cameras (monocular, stereo, RGB-D, ...)



- feature matching tends to fail during fast motion
- feature-less frames
- mono VO only estimates motion up to scale

Beyond Cameras

How to get scale and improve robustness?
add more sensors!

- wheel odometry
- GPS
- Lidar
- Inertial Measurement Unit (IMU)
- Event Cameras



830g

8 W



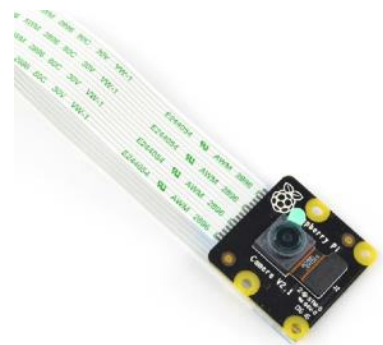
160g

2.5 W



4g

0.3W



3g

~1 W

VIO: Visual-Inertial Odometry

- (a.k.a. visual-inertial navigation, VIN)

On-Manifold Preintegration for Real-Time Visual-Inertial Odometry

Christian Forster, Luca Carlone, Frank Dellaert, Davide Scaramuzza

Abstract—Current approaches for visual-inertial odometry (VIO) are able to attain highly accurate state estimation via nonlinear optimization. However, real-time optimization quickly becomes infeasible as the trajectory grows over time; this problem is further emphasized by the fact that inertial measurements come at high rate, hence leading to fast growth of the number of variables in the optimization. In this paper, we address this issue by preintegrating inertial measurements between selected keyframes into single relative motion constraints. Our first contribution is a *preintegration theory* that properly addresses the manifold structure of the rotation group. We formally discuss the generative measurement model as well as the nature of the rotation noise and derive the expression for the *maximum a posteriori* state estimator. Our theoretical development enables the computation of all necessary Jacobians for the optimization and a-posteriori bias correction in analytic form. The second contribution is to show that the preintegrated IMU model can be seamlessly integrated into a visual-inertial pipeline under the unifying framework of factor graphs. This enables the application of incremental-smoothing algorithms and the use of a *structureless* model for visual measurements, which avoids optimizing over the 3D points, further accelerating the computation. We perform an extensive evaluation of our monocular VIO pipeline on real and simulated datasets. The results confirm that our modelling effort leads to accurate state estimation in real-time, outperforming state-of-the-art approaches.

of monocular vision and gravity observable [1] and provides robust and accurate inter-frame motion estimates. Applications of VIO range from autonomous navigation in GPS-denied environments, to 3D reconstruction, and augmented reality.

The existing literature on VIO imposes a trade-off between accuracy and computational efficiency (a detailed review is given in Section II). On the one hand, filtering approaches enable fast inference, but their accuracy is deteriorated by the accumulation of linearization errors. On the other hand, full smoothing approaches, based on nonlinear optimization, are accurate, but computationally demanding. Fixed-lag smoothing offers a compromise between accuracy for efficiency; however, it is not clear how to set the length of the estimation window so to guarantee a given level of performance.

In this work we show that it is possible to overcome this trade-off. We design a VIO system that enables fast incremental smoothing and computes the optimal *maximum a posteriori* (MAP) estimate in real time. An overview of our approach is given in Section IV.

The first step towards this goal is the development of a novel preintegration theory. The use of *preintegrated IMU measurements* was first proposed in [2] and consists of combining

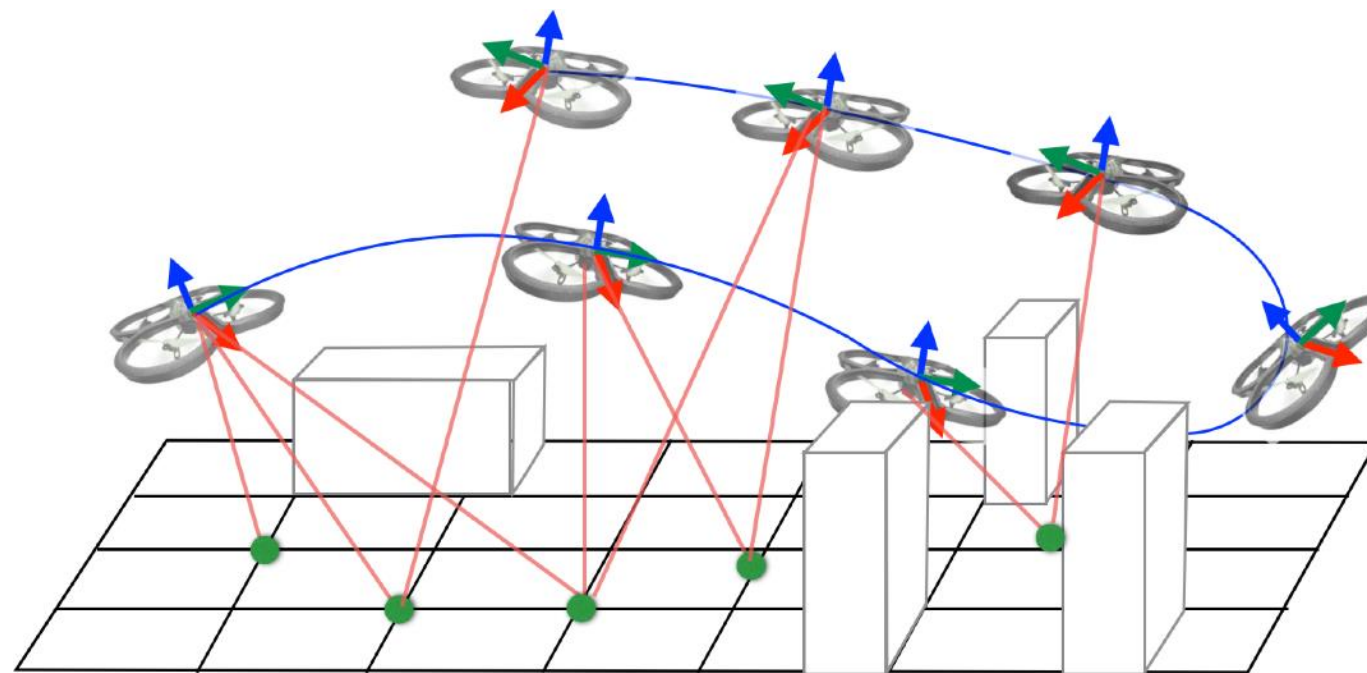
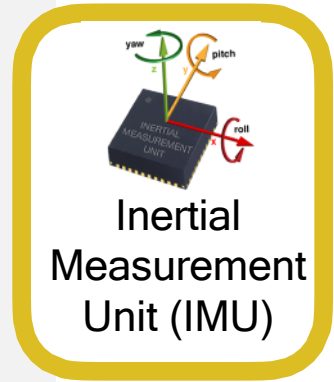
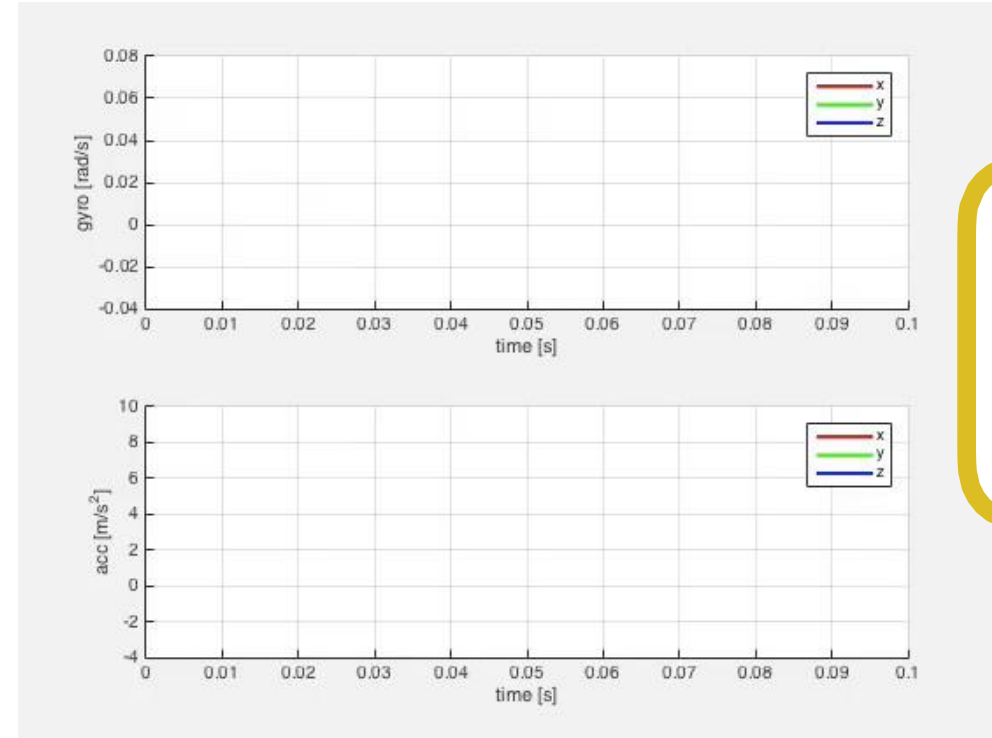
More resources:

Visual-Inertial Navigation: Challenges and Applications

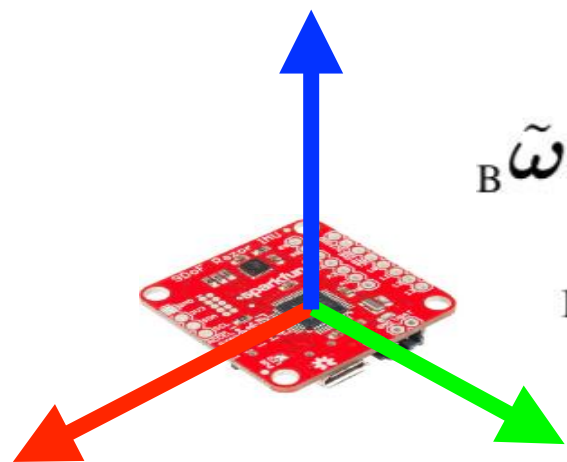
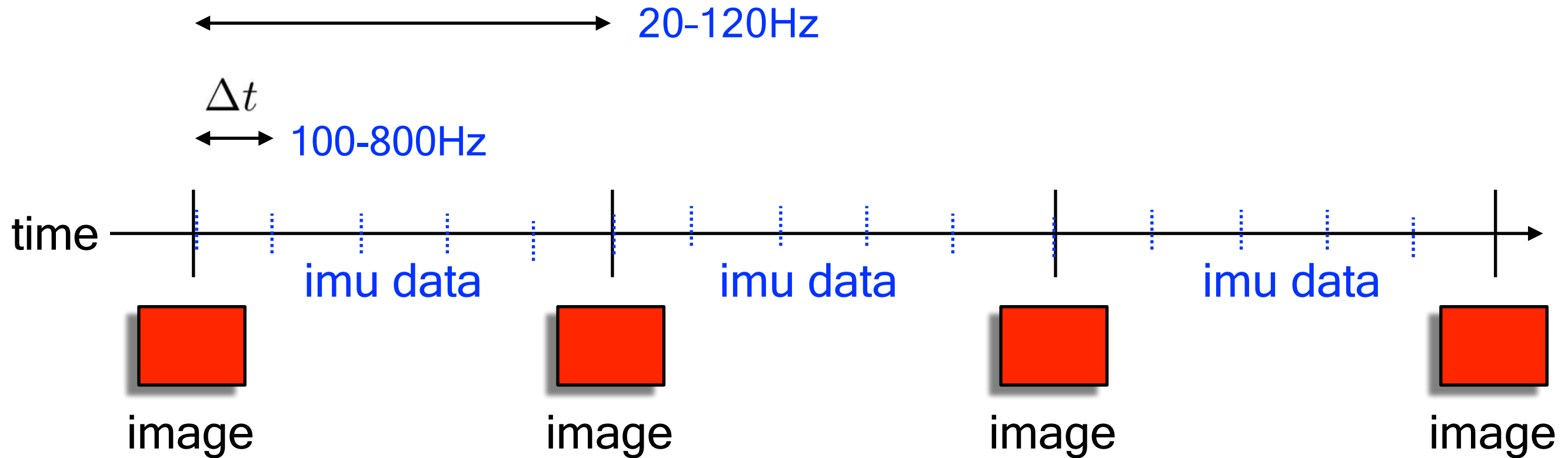
IROS 2019 Full-day Workshop: November 8, 2019, Macau, China

<https://udel.edu/~ghuang/iros19-vins-workshop/>

Visual-Inertial Navigation (VIN)

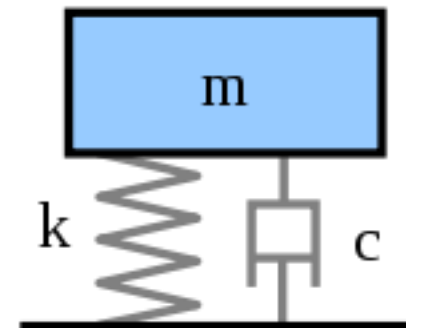


Visual-Inertial Odometry

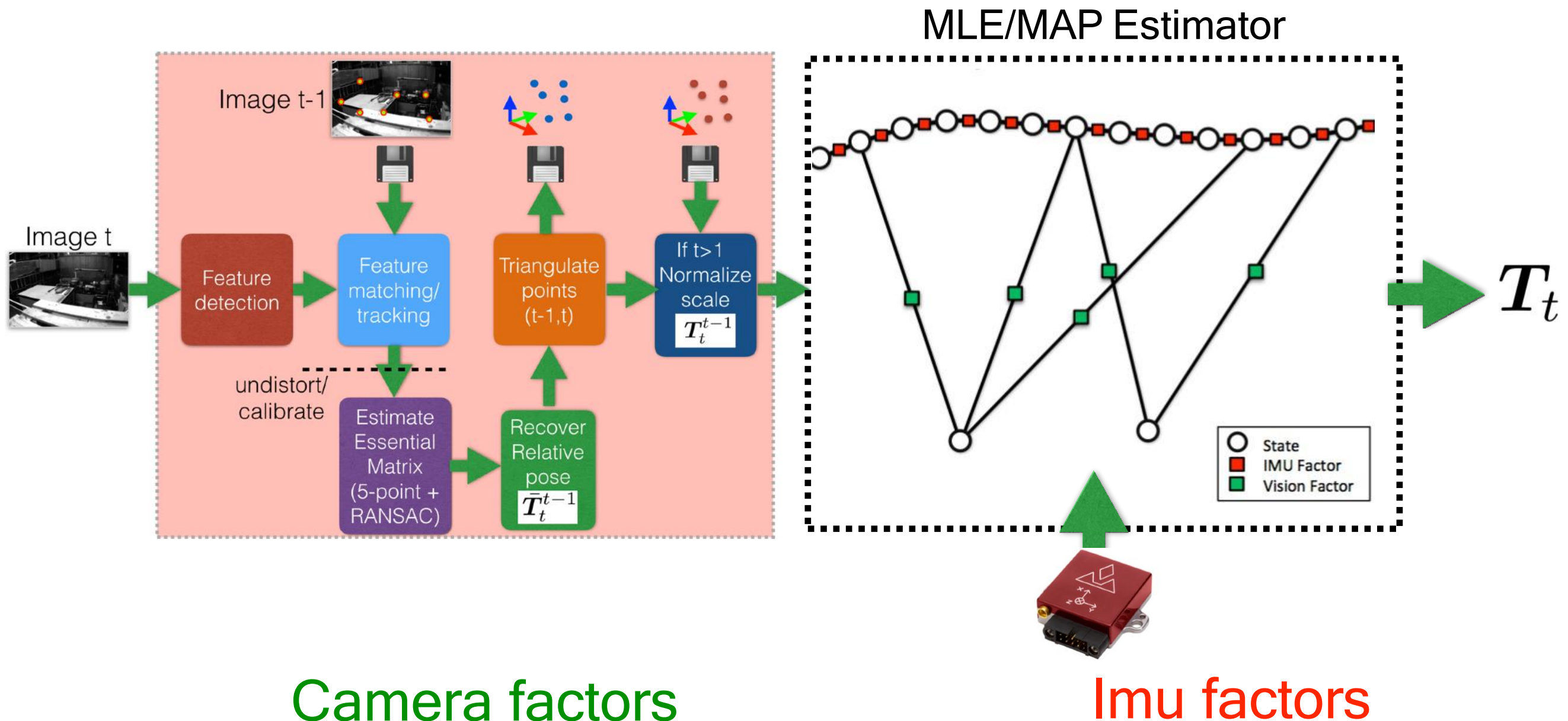


$${}_{\text{B}}\tilde{\boldsymbol{\omega}}_{\text{WB}}(t) = {}_{\text{B}}\boldsymbol{\omega}_{\text{WB}}(t) + \mathbf{b}^g(t) + \boldsymbol{\eta}^g(t)$$

$${}_{\text{B}}\tilde{\mathbf{a}}(t) = \mathbf{R}_{\text{WB}}^{\text{T}}(t) ({}_{\text{W}}\mathbf{a}(t) - {}_{\text{W}}\mathbf{g}) + \mathbf{b}^a(t) + \boldsymbol{\eta}^a(t),$$



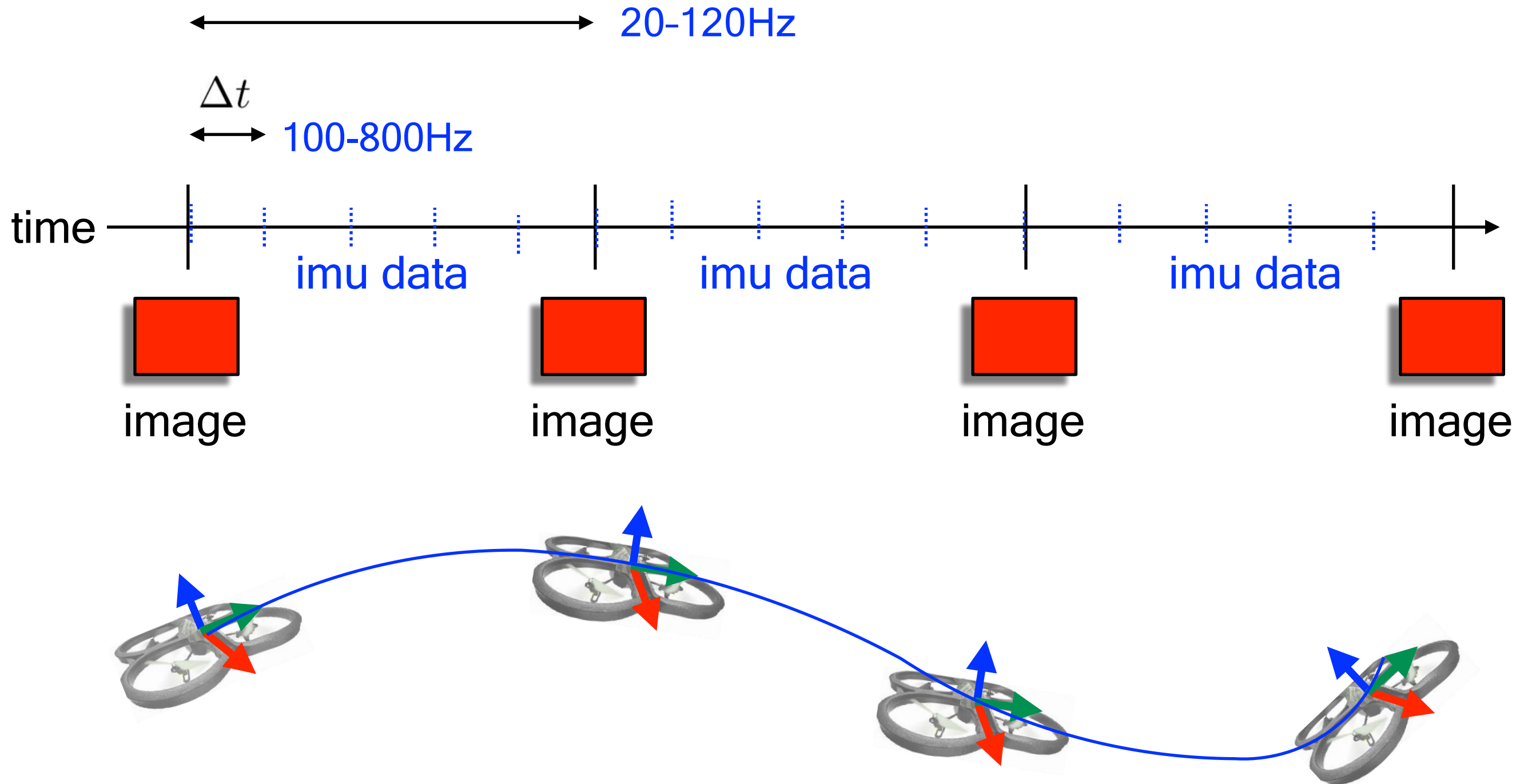
Visual-Inertial Odometry



$$\min_{\mathbf{T}_i, i=1, \dots, N_C} \sum_{k=1}^N \sum_{i \in \mathcal{C}_k} \|\mathbf{x}_{k,i} - \pi(\mathbf{T}_i, \mathbf{p}_k)\|^2 + \sum_{i=1, \dots, N_C-1} \|r_{\text{imu}}(\mathbf{T}_i, \mathbf{T}_{i+1}, \mathbf{v}_i, \mathbf{v}_{i+1}, \mathbf{b}_i, \mathbf{b}_{i+1})\|^2$$

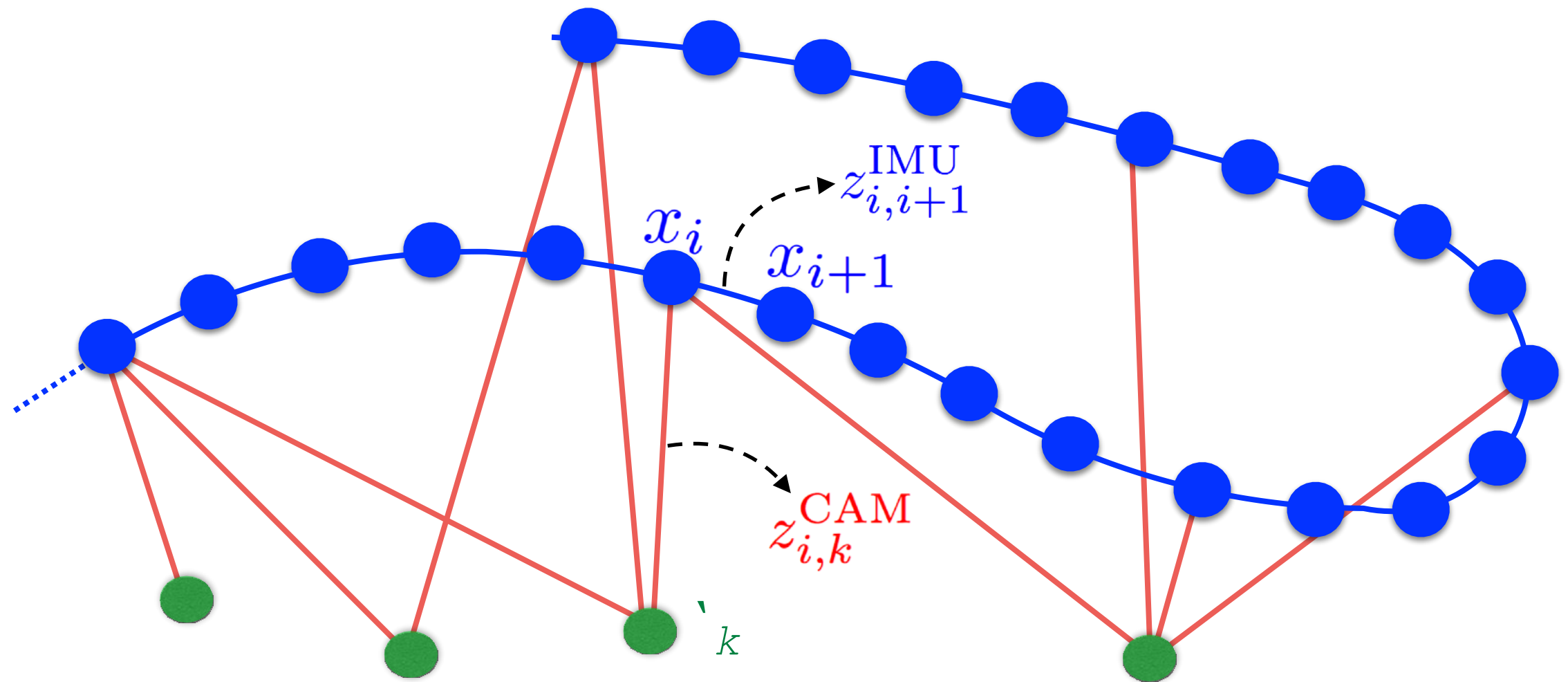
Need to include velocities and IMU biases (gyroscopes and accelerometers) in the state ...

Visual-Inertial Odometry



- **Fixed-lag smoother:** estimate a fixed window of recent states from time $k-T$, $k-T+1$, .. k (sliding window)

MAP Estimation



Challenges:

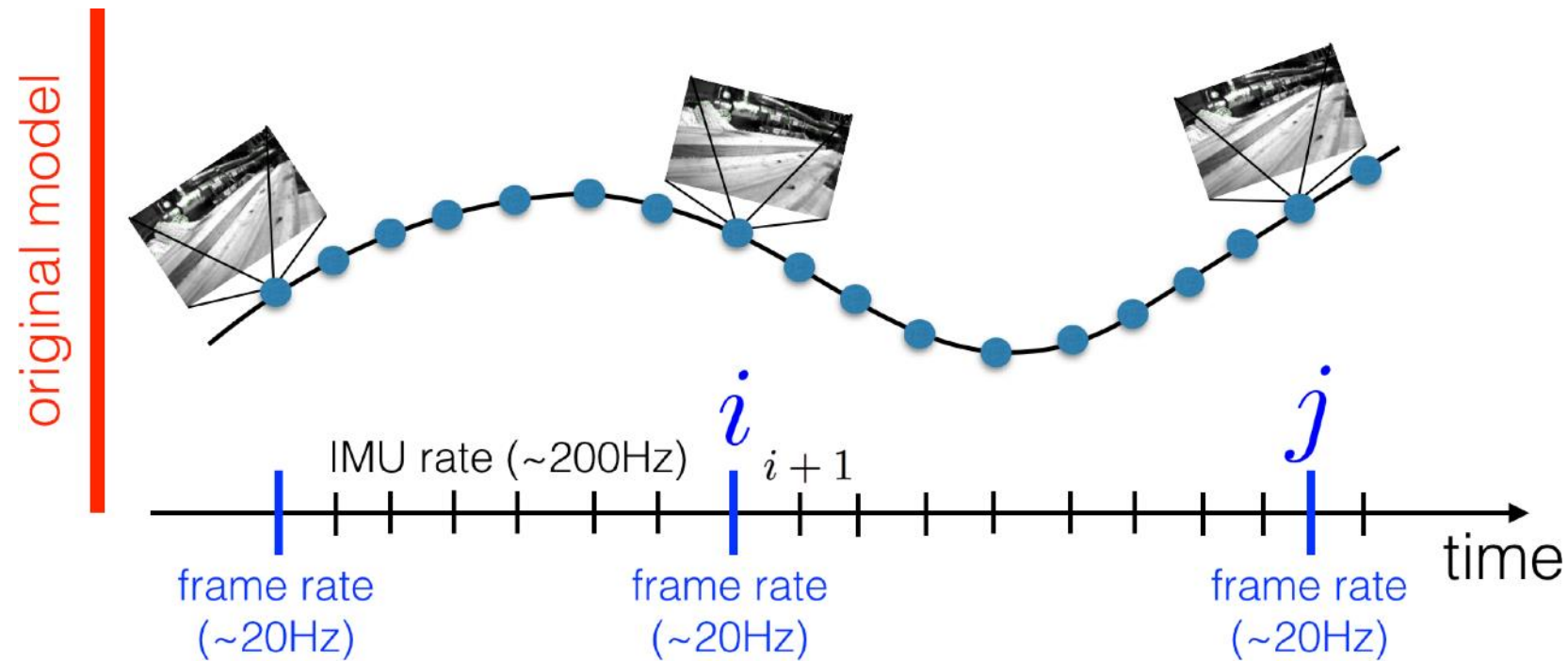
- IMU measurements arrive at high-rate ($\sim 200\text{Hz}$) ➡ IMU preintegration
- camera observes hundreds of landmarks per frame ➡ structureless vision factors
- need to solve optimization problem quickly

IMU Preintegration

Key idea: integrate IMU measurements between frames

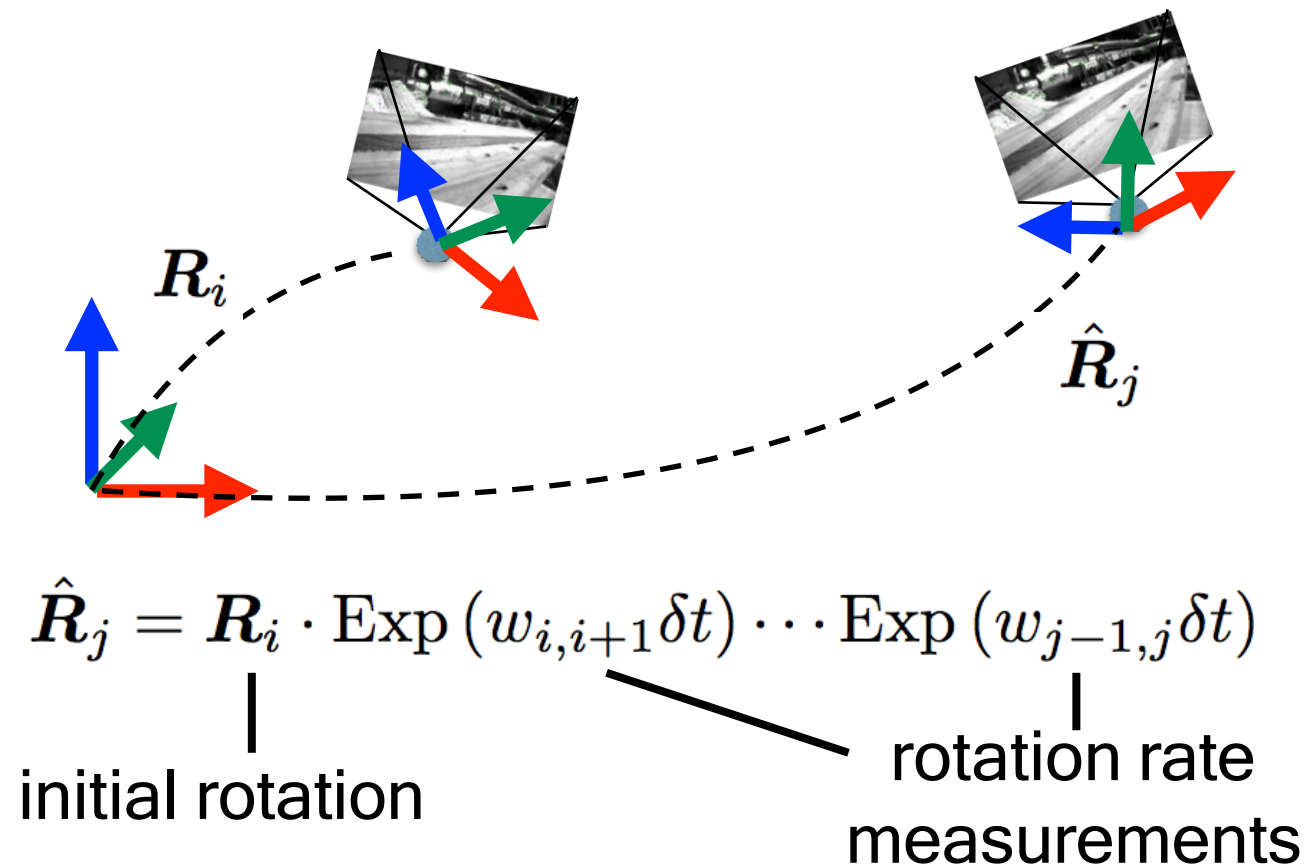
many measurements & states

$$\begin{aligned} z_{i,i+1}^{\text{IMU}} &= f(x_i, x_{i+1}) + \epsilon \\ z_{i+1,i+2}^{\text{IMU}} &= f(x_{i+1}, x_{i+2}) + \epsilon \\ &\vdots \\ z_{j-1,j}^{\text{IMU}} &= f(x_{j-1}, x_j) + \epsilon \end{aligned}$$

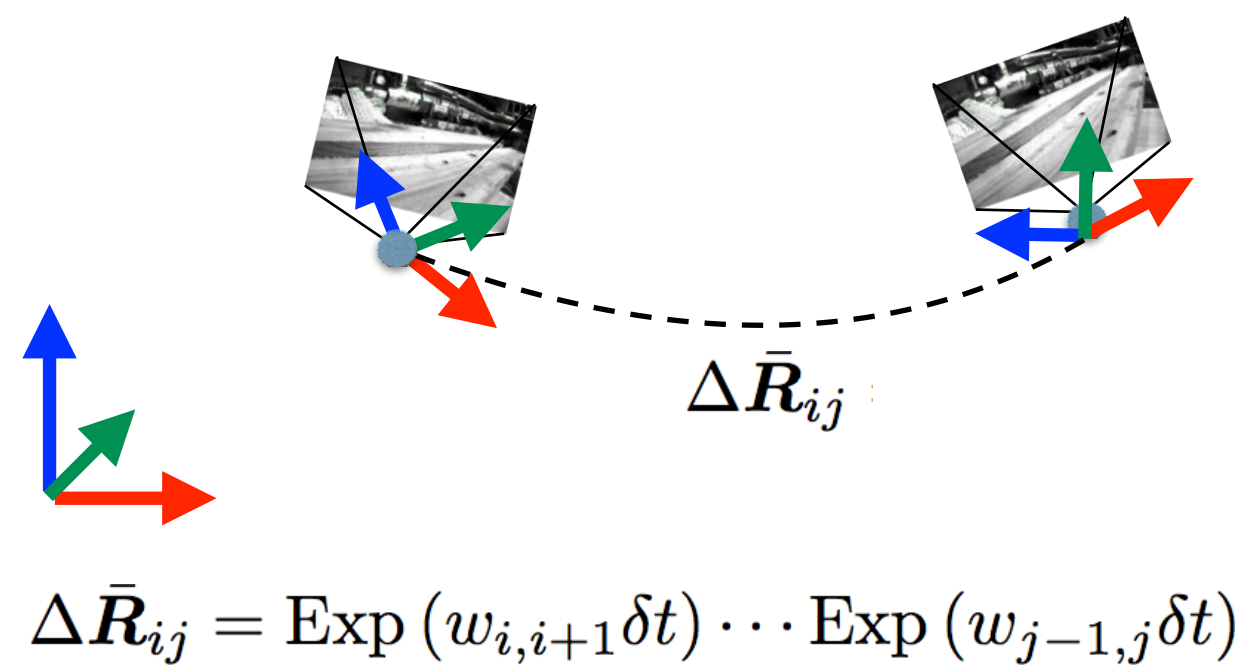


IMU Preintegration

Standard integration

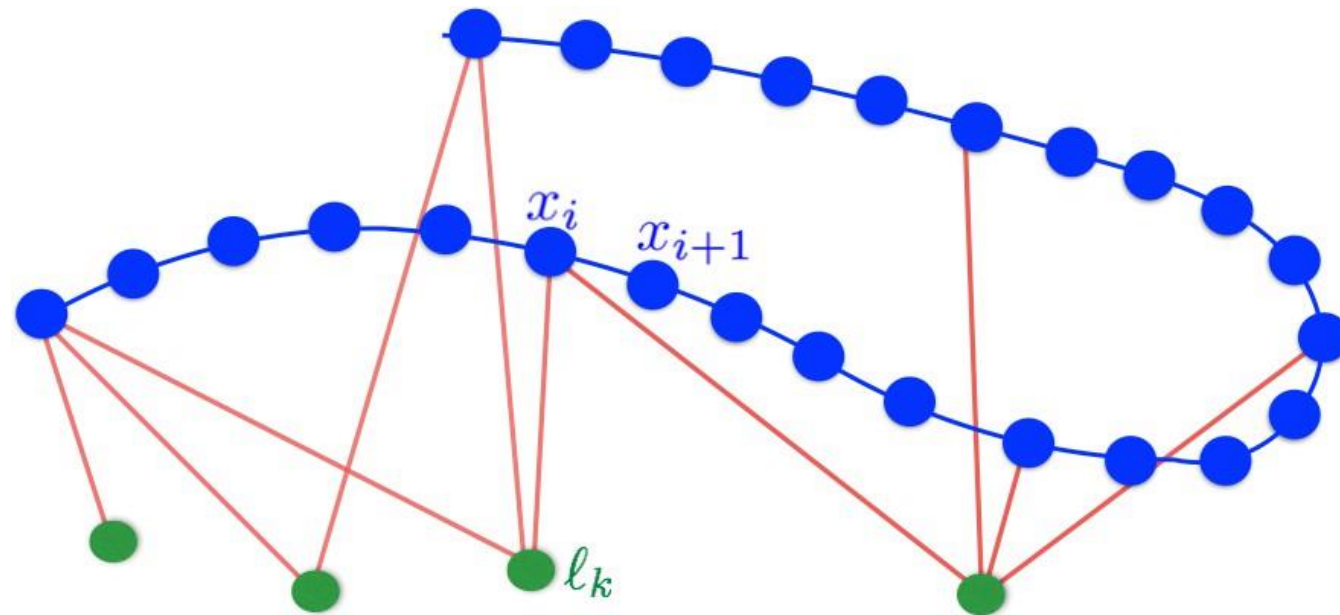


Preintegration



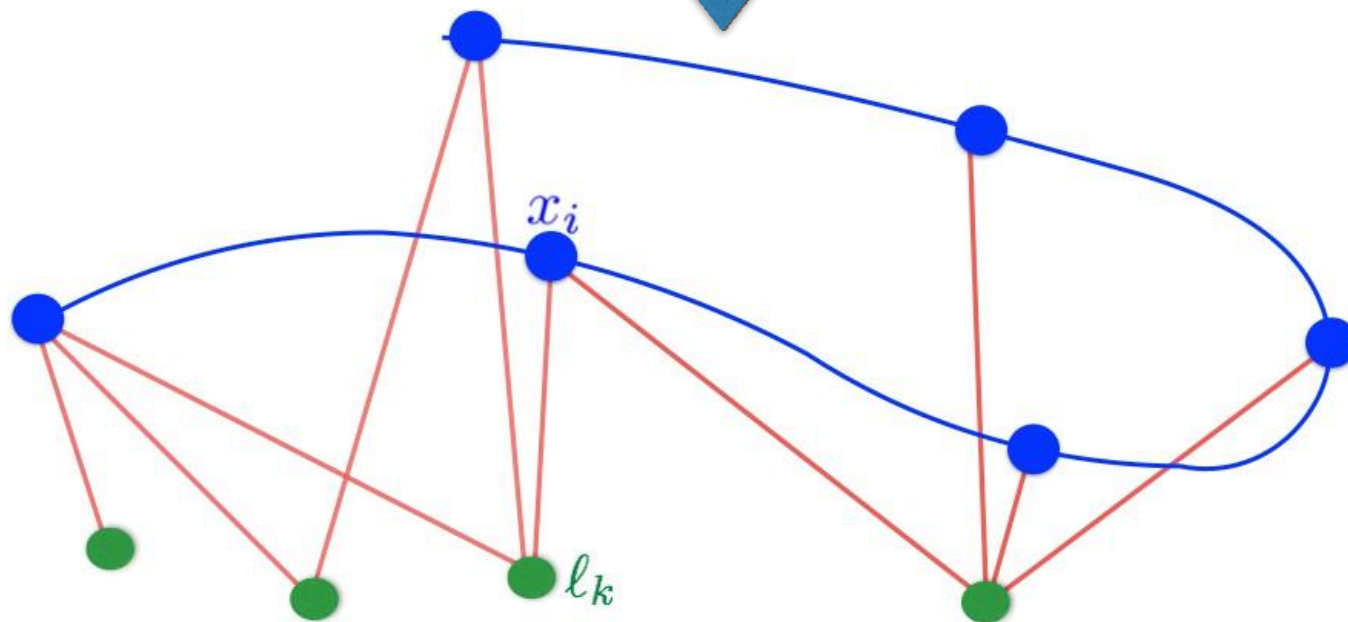
The preintegration allows us to accurately summarize hundreds of inertial measurements into a single relative motion constraint.

IMU Preintegration



After 10 seconds, original problem has $\sim 10^4$ states

Preintegration

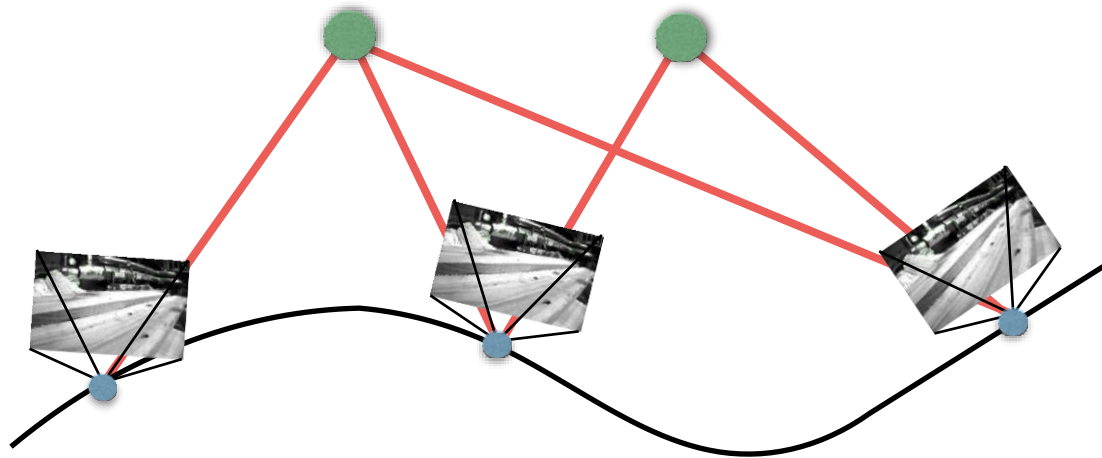


After 10 seconds, preintegrated problem has $\sim 10^2$ states

On-Manifold Preintegration for Real-Time Visual-Inertial Odometry

Structureless Vision Model

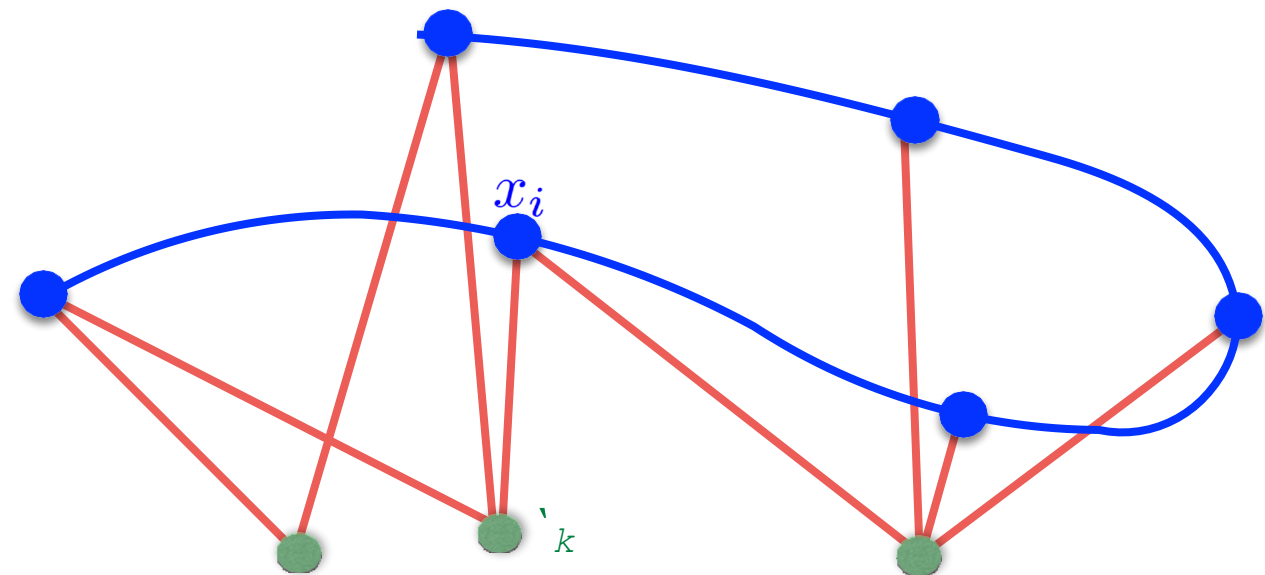
Marginalization of 3D landmarks



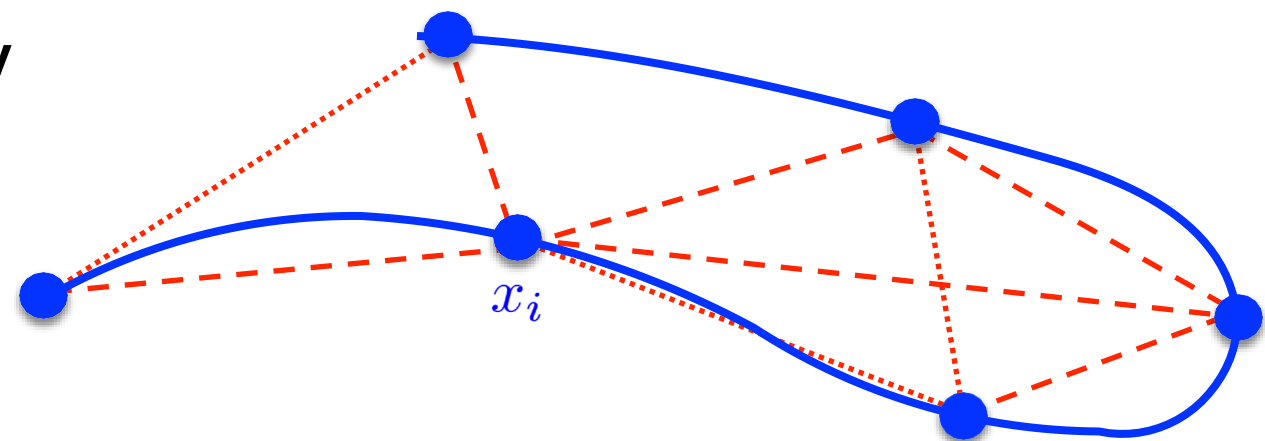
Schur complement trick:

- solve for each landmark separately
- substitute back in the optimization

Further reduction of the number
of variables in the optimization!

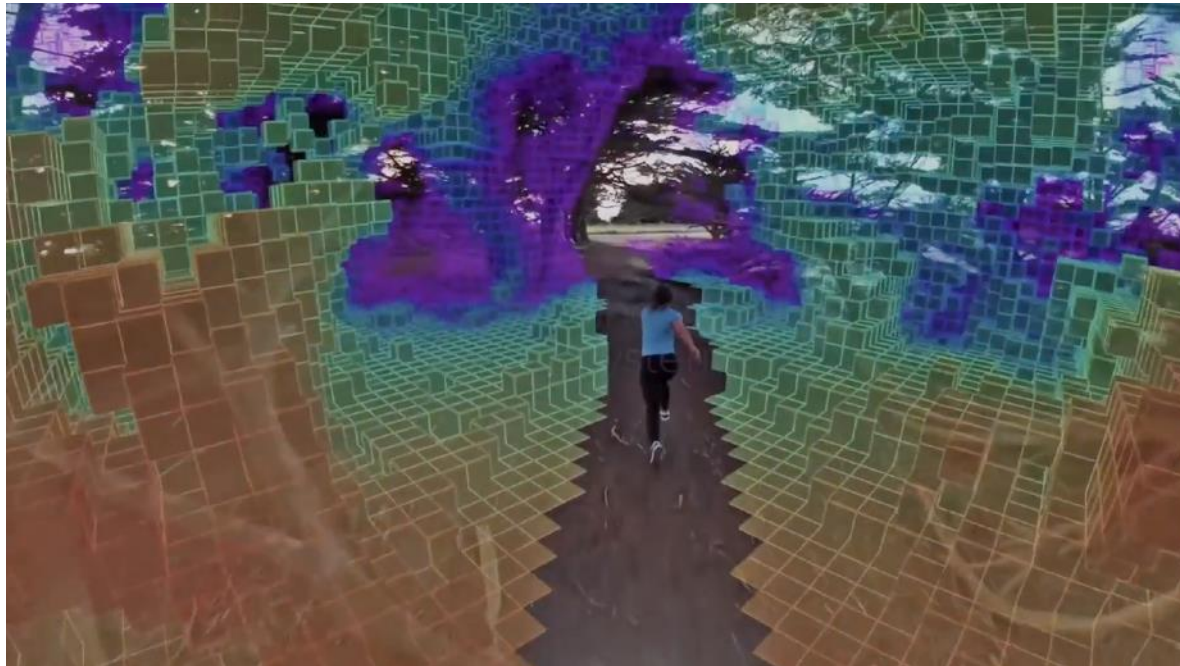


Schur complement

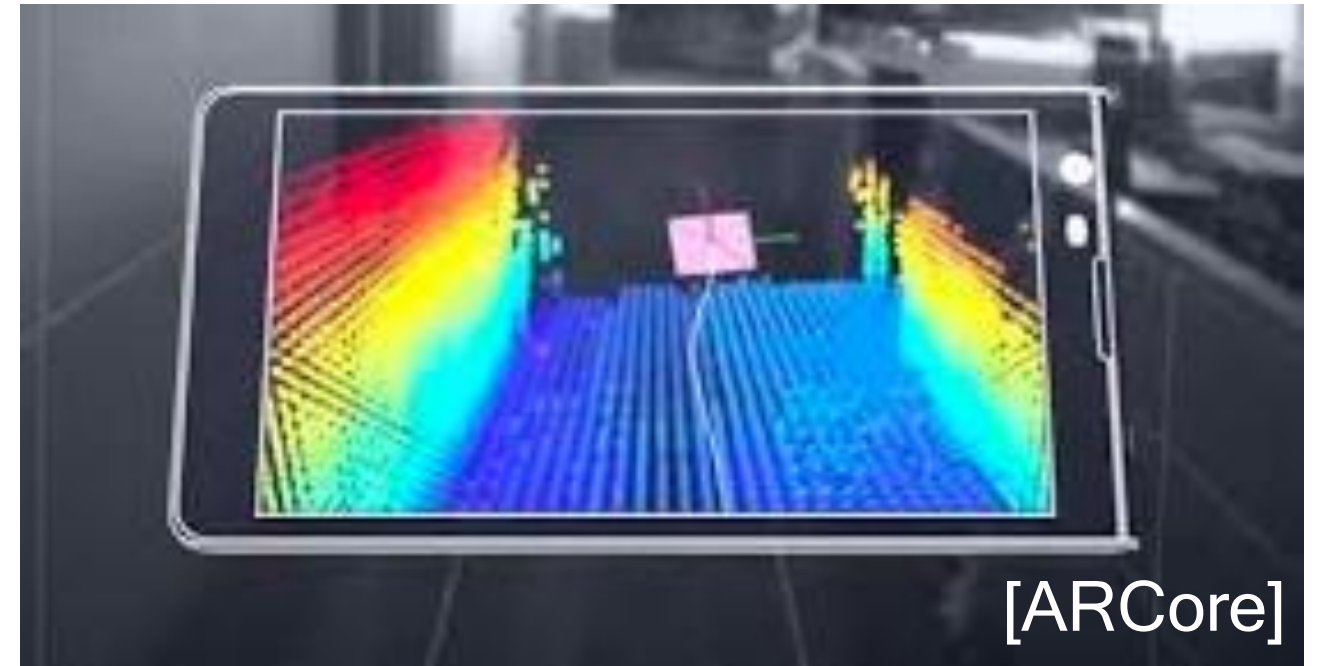


Engineered Solutions / Applications

Skydio R1 drone



Google Tango



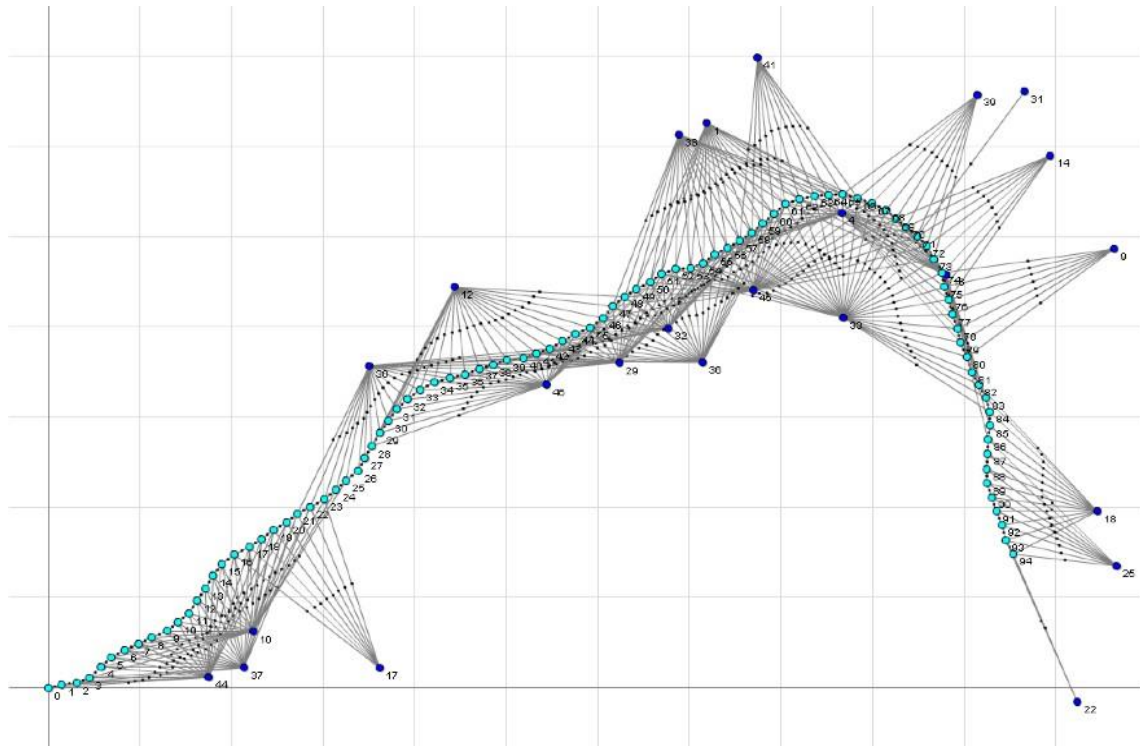
Oculus Rift Goggles



Pokemon Go



SLAM



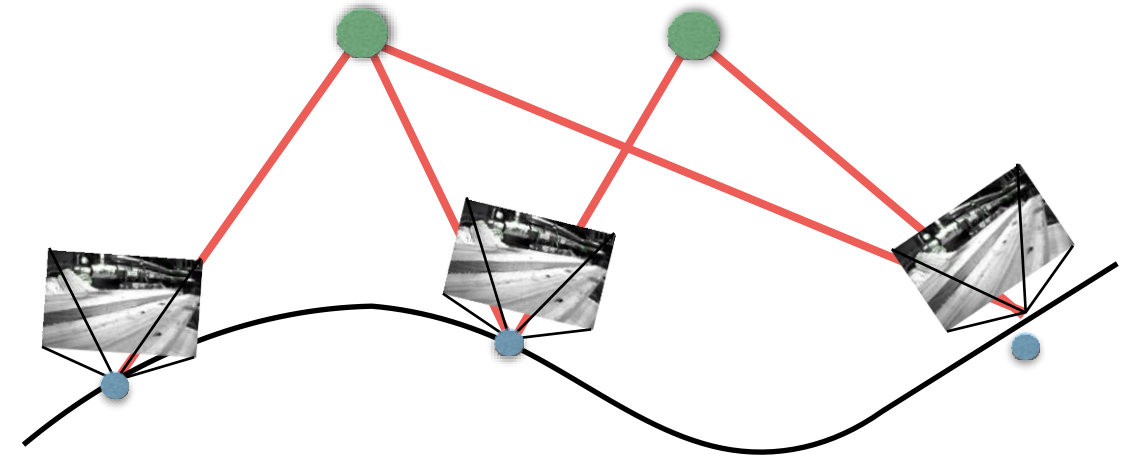
All sensors

Optimize entire history
(full smoothing)

Latency:
>200ms

Loop closures: error
remains bounded

VIN



Camera & IMU

Typically optimize
over receding horizon

Latency:
<50ms

No loop closures: error
accumulates over time

Beyond Cameras

- wheel odometry
- GPS
- Lidar
- Inertial Measurement Unit (IMU)
- Event Cameras



830g

8 W



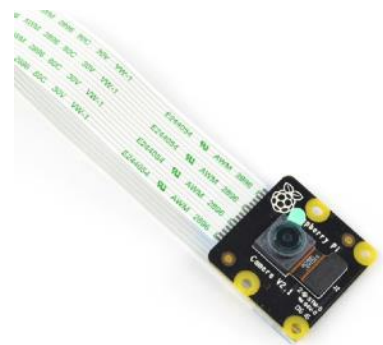
160g

2.5 W



4g

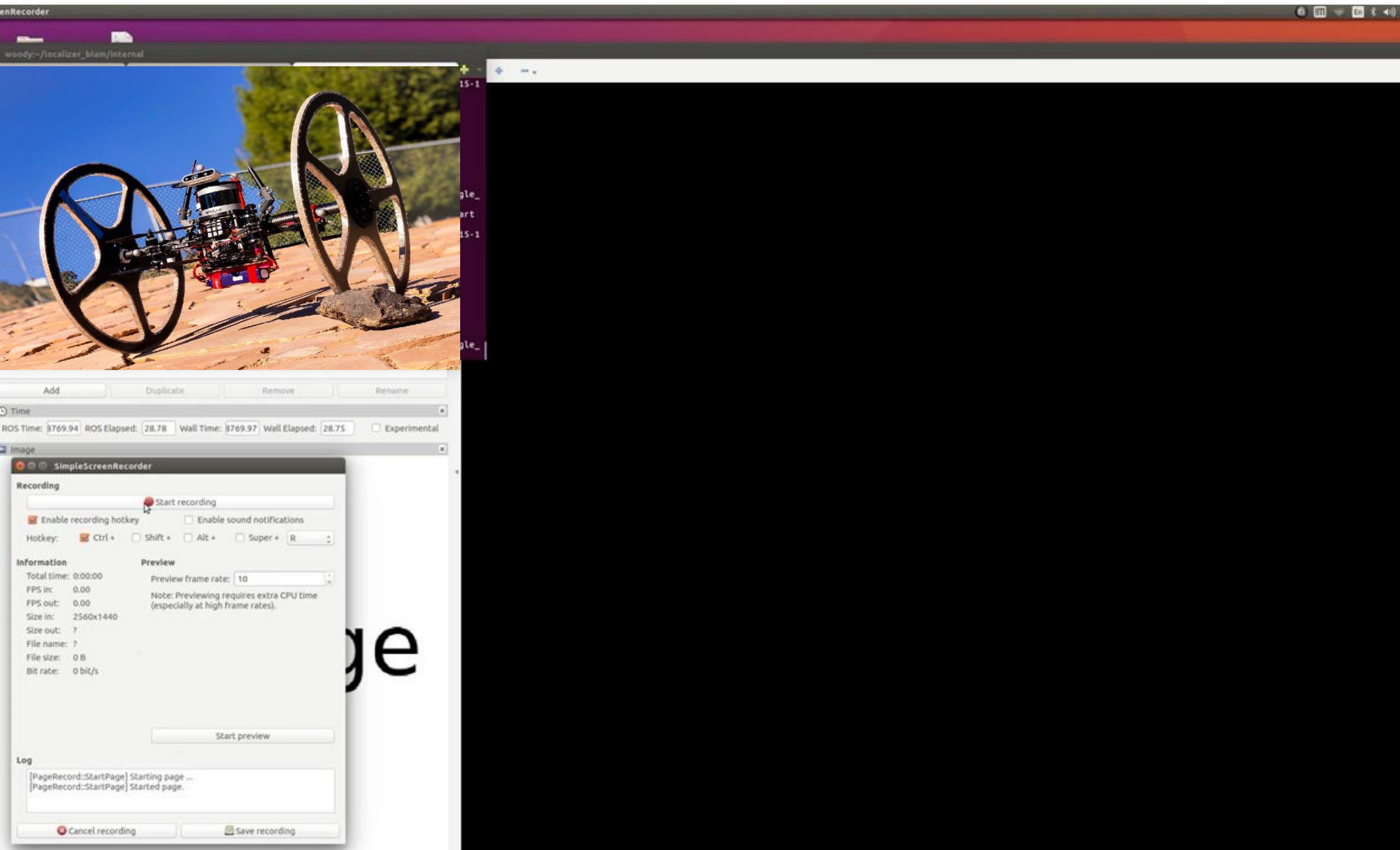
0.3W



3g

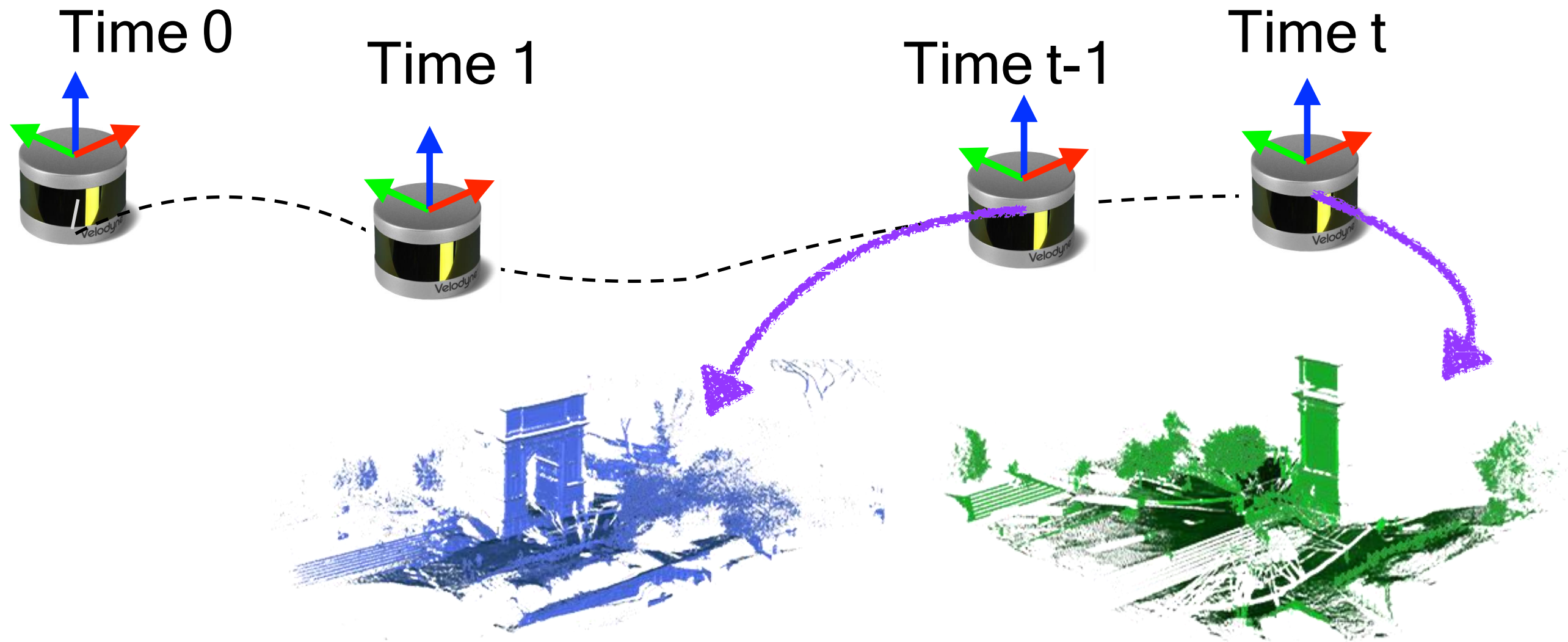
~1 W

Lidar Odometry & Lidar SLAM

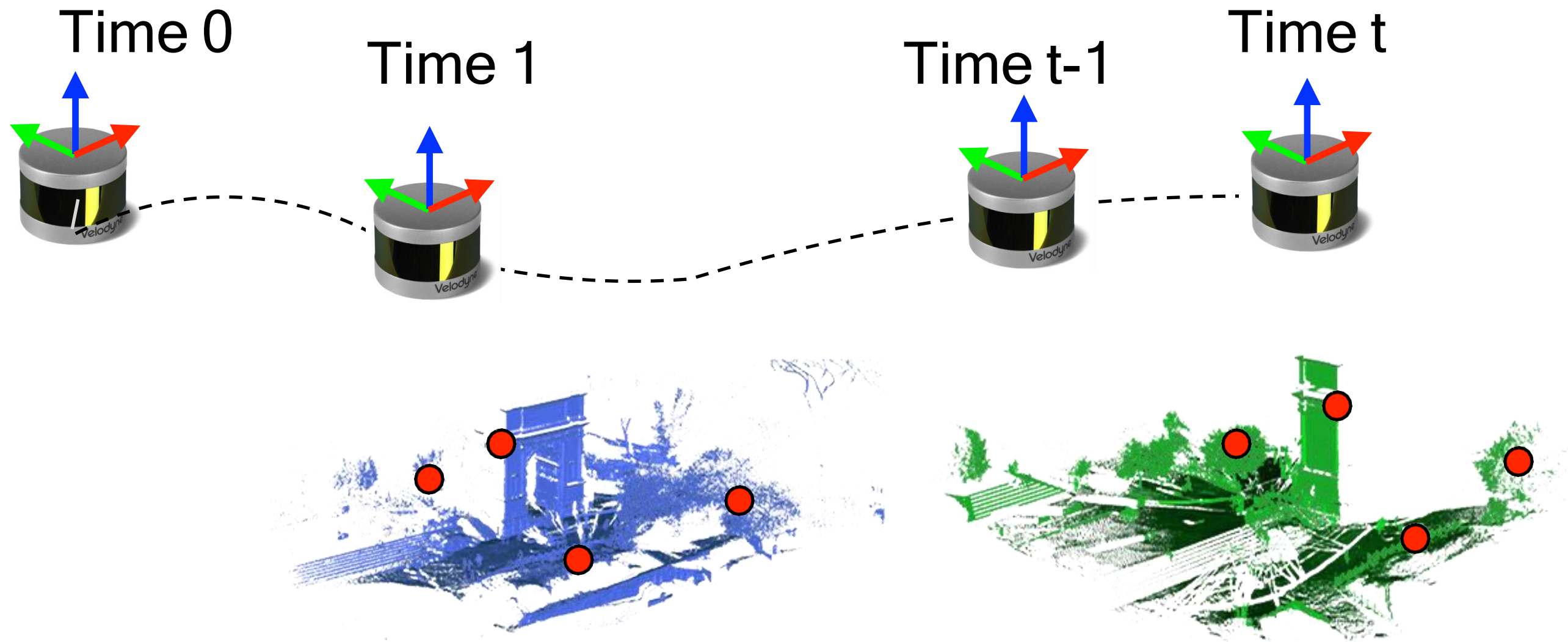


DARPA Subterranean Challenge, in collaboration with JPL

Feature-based Lidar Odometry



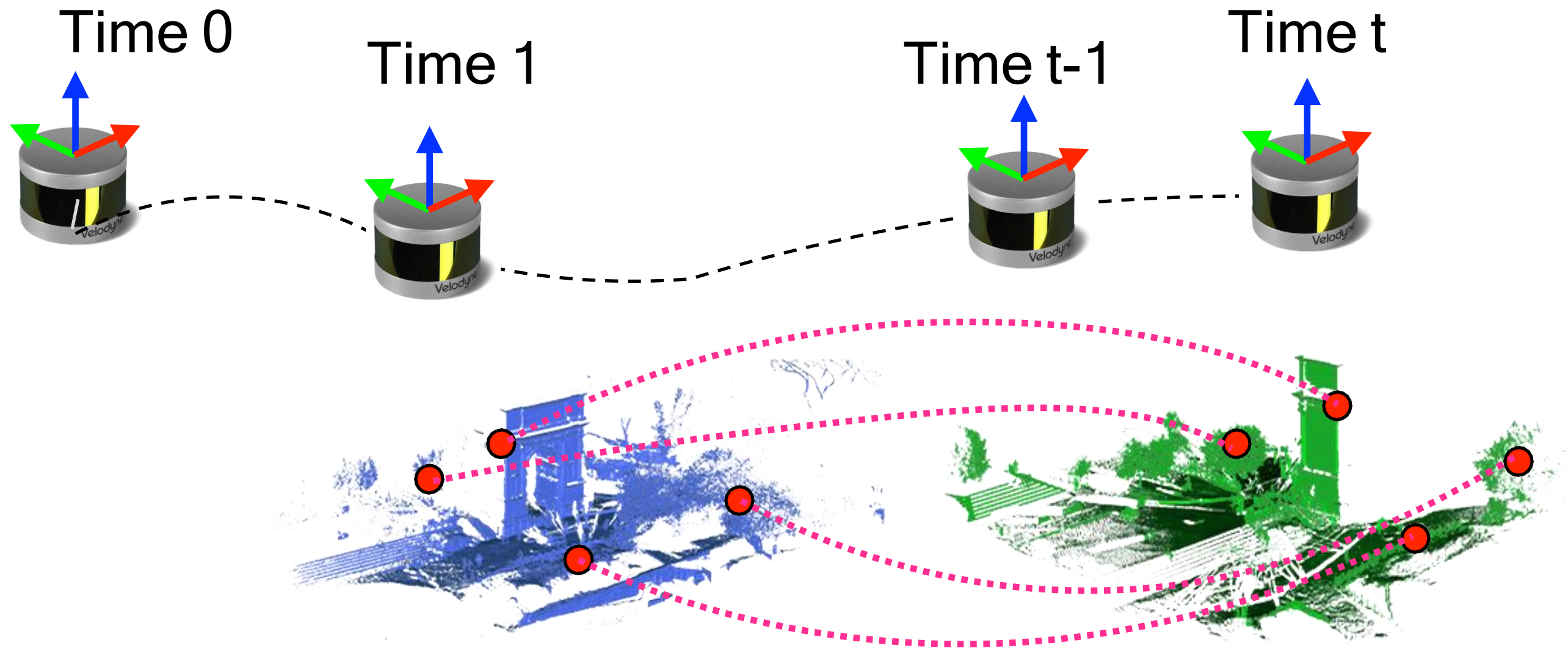
Feature-based Lidar Odometry



Registration: compute relative pose between scans:

- extract features & descriptors
- use descriptors for matching
- compute relative pose

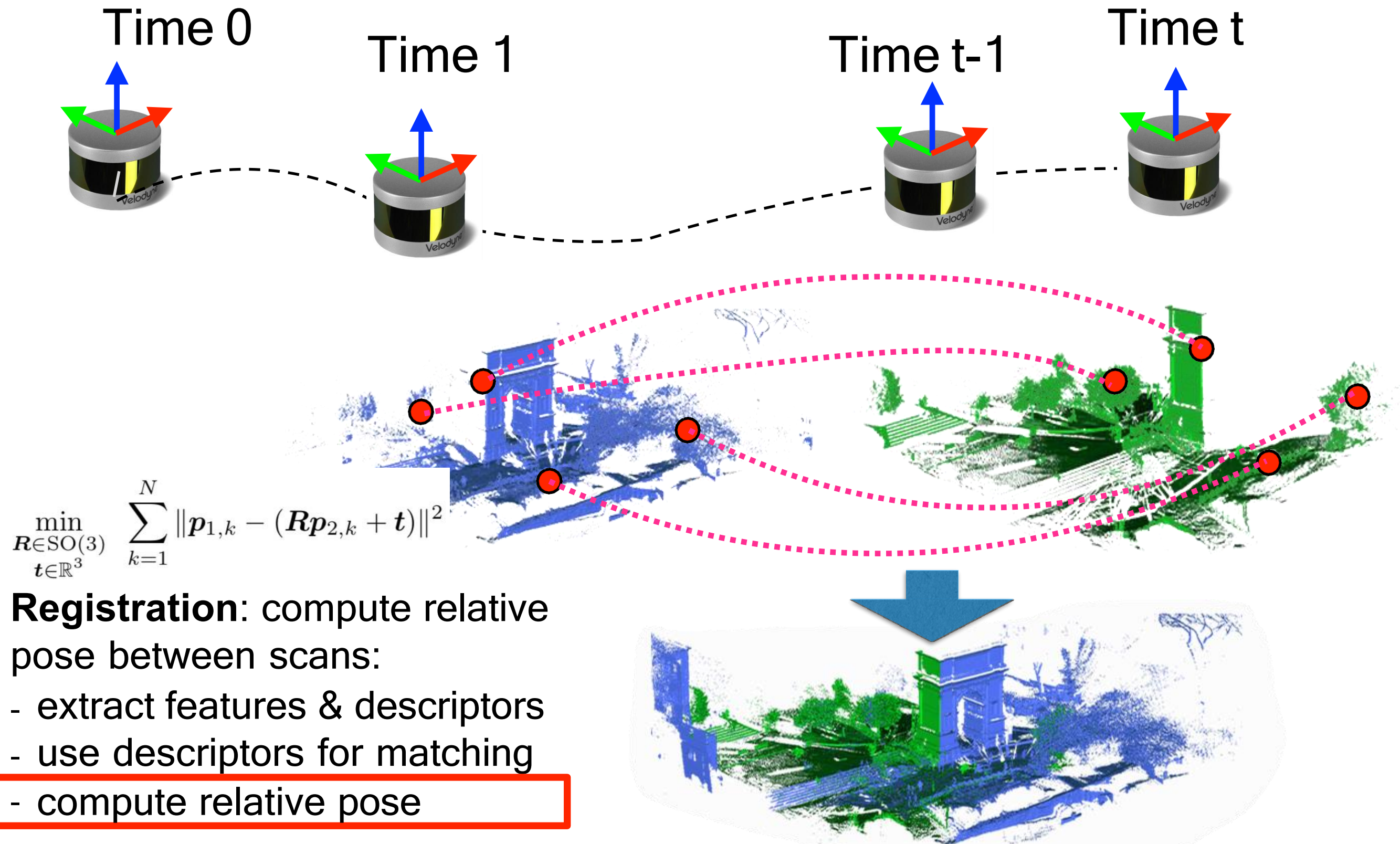
Feature-based Lidar Odometry



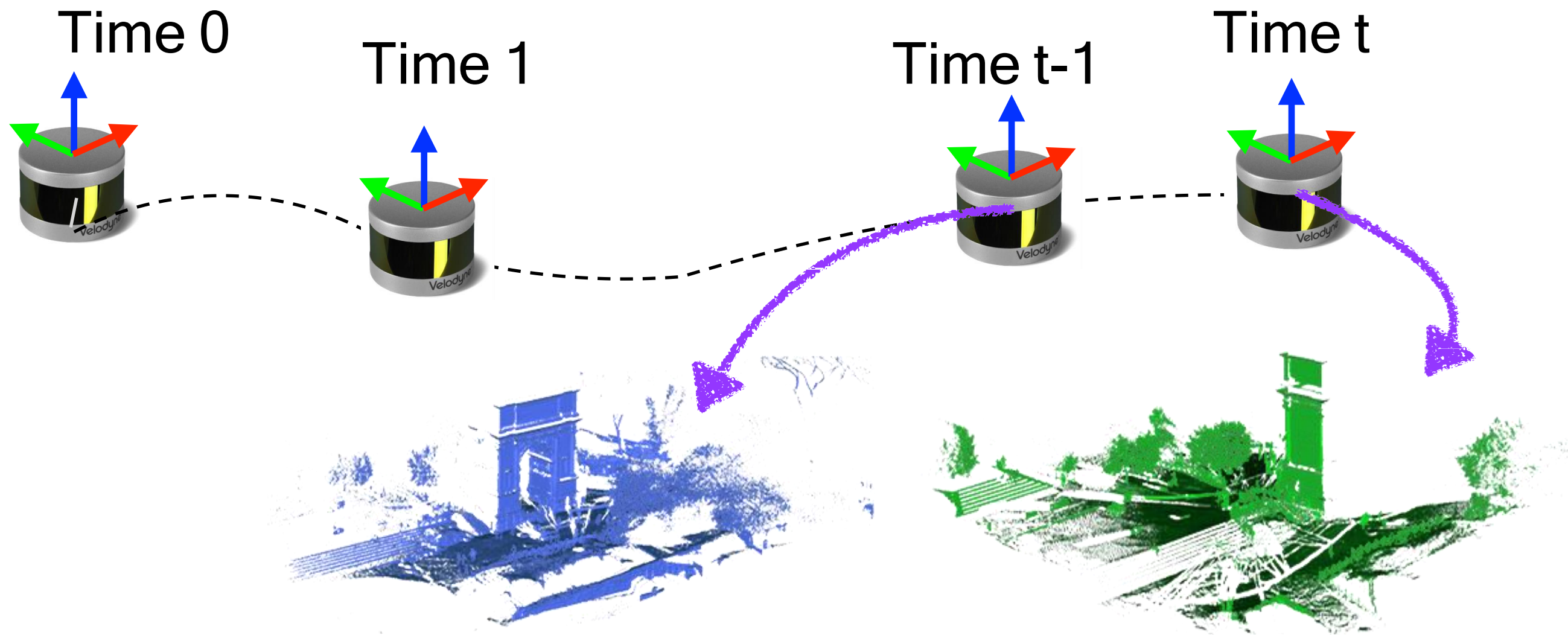
Registration: compute relative pose between scans:

- extract features & descriptors
- use descriptors for matching
- compute relative pose

Feature-based Lidar Odometry



Dense Lidar Odometry



Iterative Closest Point (ICP)

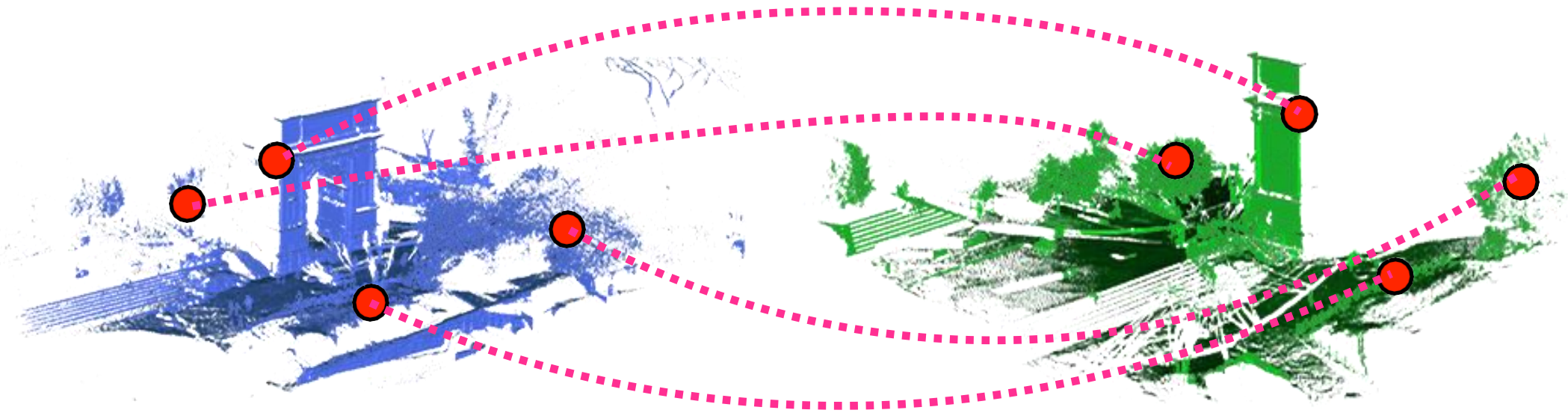
- Alternative to feature-based approaches
- Simultaneous Pose and Correspondences

Iterative Closest Point (ICP)

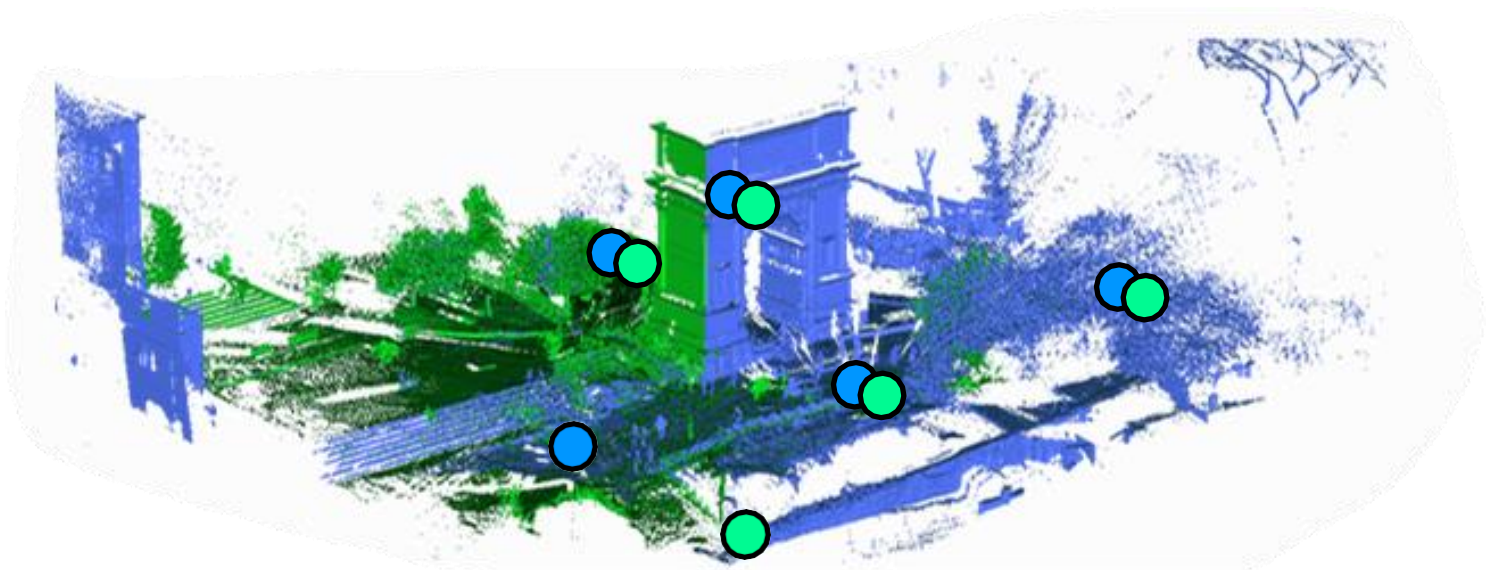
- **Observations:**

$$\min_{\substack{R \in \text{SO}(3) \\ t \in \mathbb{R}^3}} \sum_{k=1}^N \|p_{1,k} - (Rp_{2,k} + t)\|^2$$

1. Easy to compute alignment given ground-truth correspondences



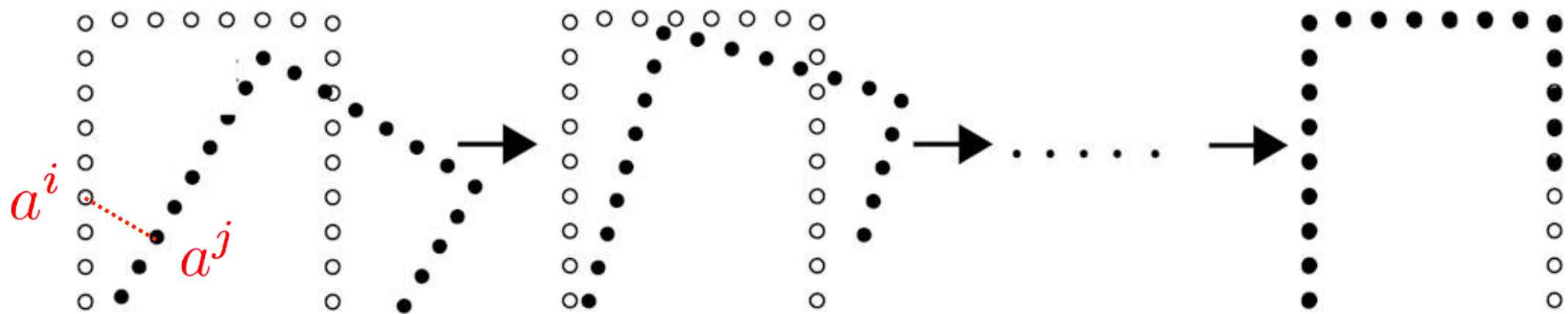
2. Easy to compute correspondences given ground-truth alignment



Iterative Closest Point (ICP)

ICP algorithm: given initial guess, perform the following:

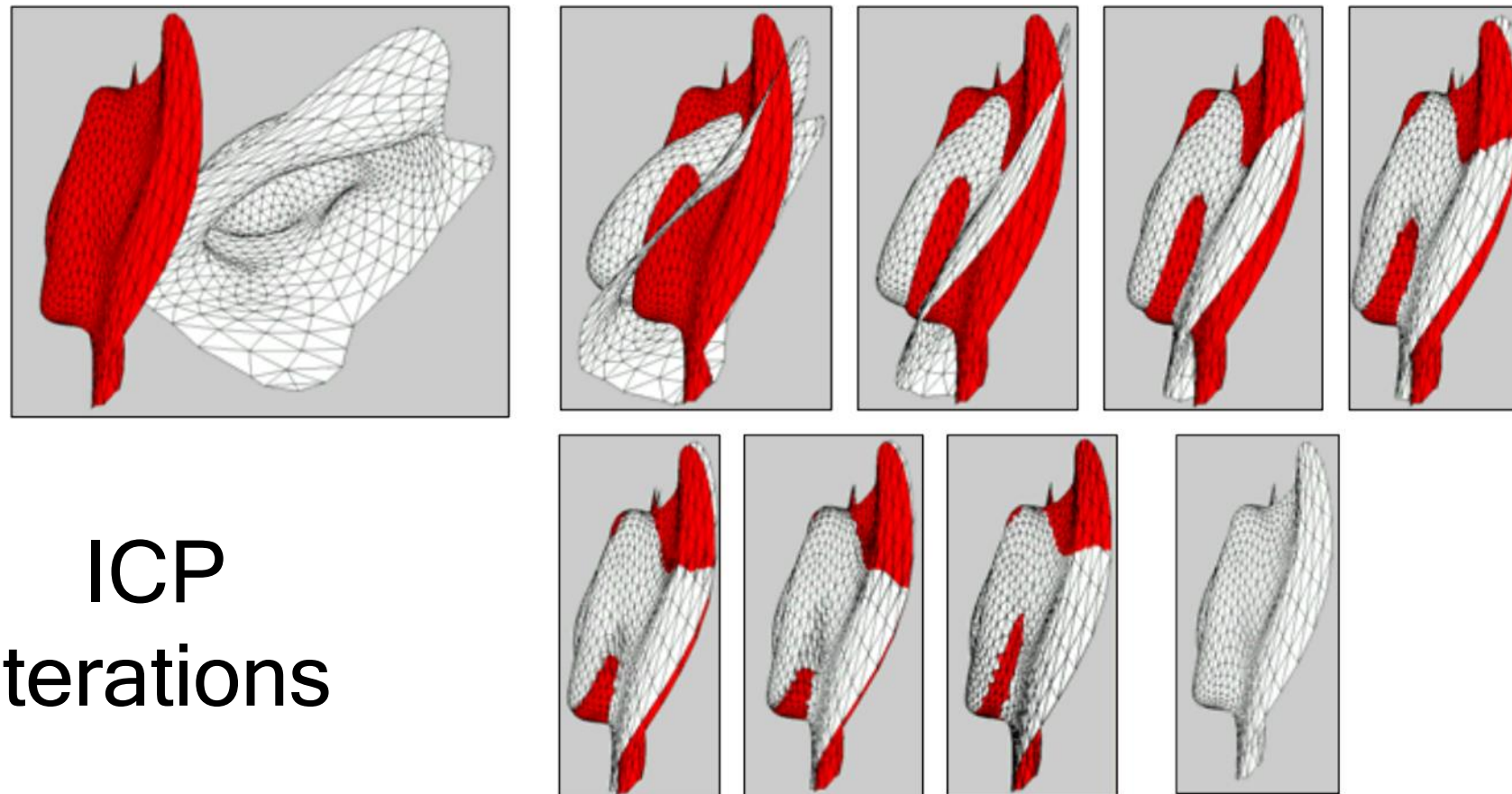
- 1. Establish correspondences:** associate to each point in Cloud 1 the closest point in Cloud 2
- 2. Compute relative pose given correspondences**
(e.g., using Horn's or Arun's method)
- 3. Transform point cloud and repeat**
(stop when alignment does not improve or after max iter.)



Iterative Closest Point (ICP)

ICP algorithm: given initial guess, perform the following:

- 1. Establish correspondences:** associate to each point in Cloud 1 the closest point in Cloud 2
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(e.g., using Horn's or Arun's method)
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ICP
Iterations

[courtesy: <http://www.cs.technion.ac.il/~cs236329/tutorials/ICP.pdf>]

Beyond Cameras

- wheel odometry
- GPS
- Lidar
- Inertial Measurement Unit (IMU)
- Event Cameras



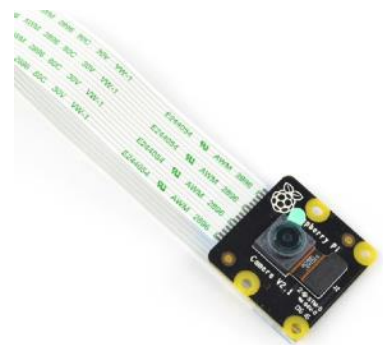
830g



160g



4g



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8 W

2.5 W

0.3W

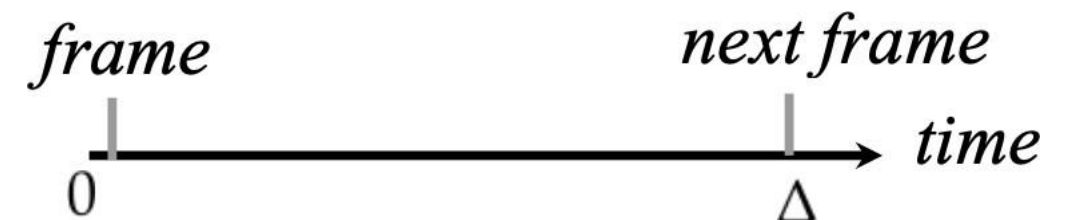
~1 W

Event-based Vision: A Survey

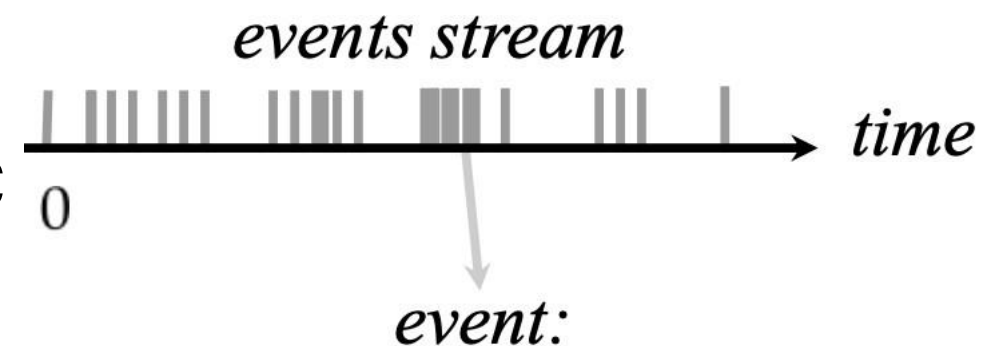
Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J. Davison, Jörg Conradt, Kostas Daniilidis, Davide Scaramuzza

Event-based Cameras

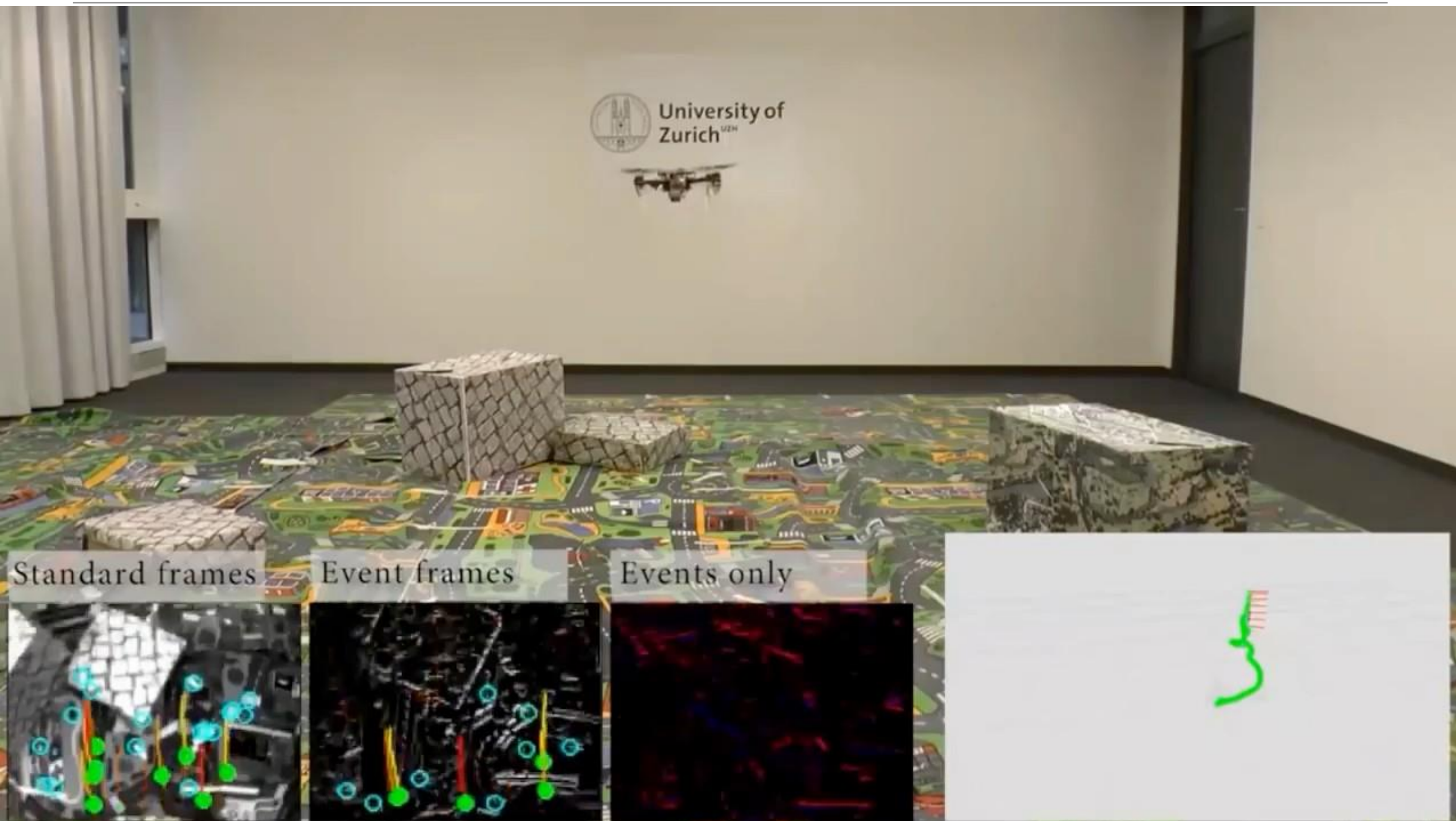
- Speed of robot is constrained by speed at which it can sense (and compute)
- Common cameras: 20-120fps



- event-based cameras (e.g., Dynamic Vision Sensor, DVS)
 - Temporal resolution: 1microsec
 - High dynamic range: 120 dB
 - Low power: 20 mW
 - Cost: 2,500 EUR

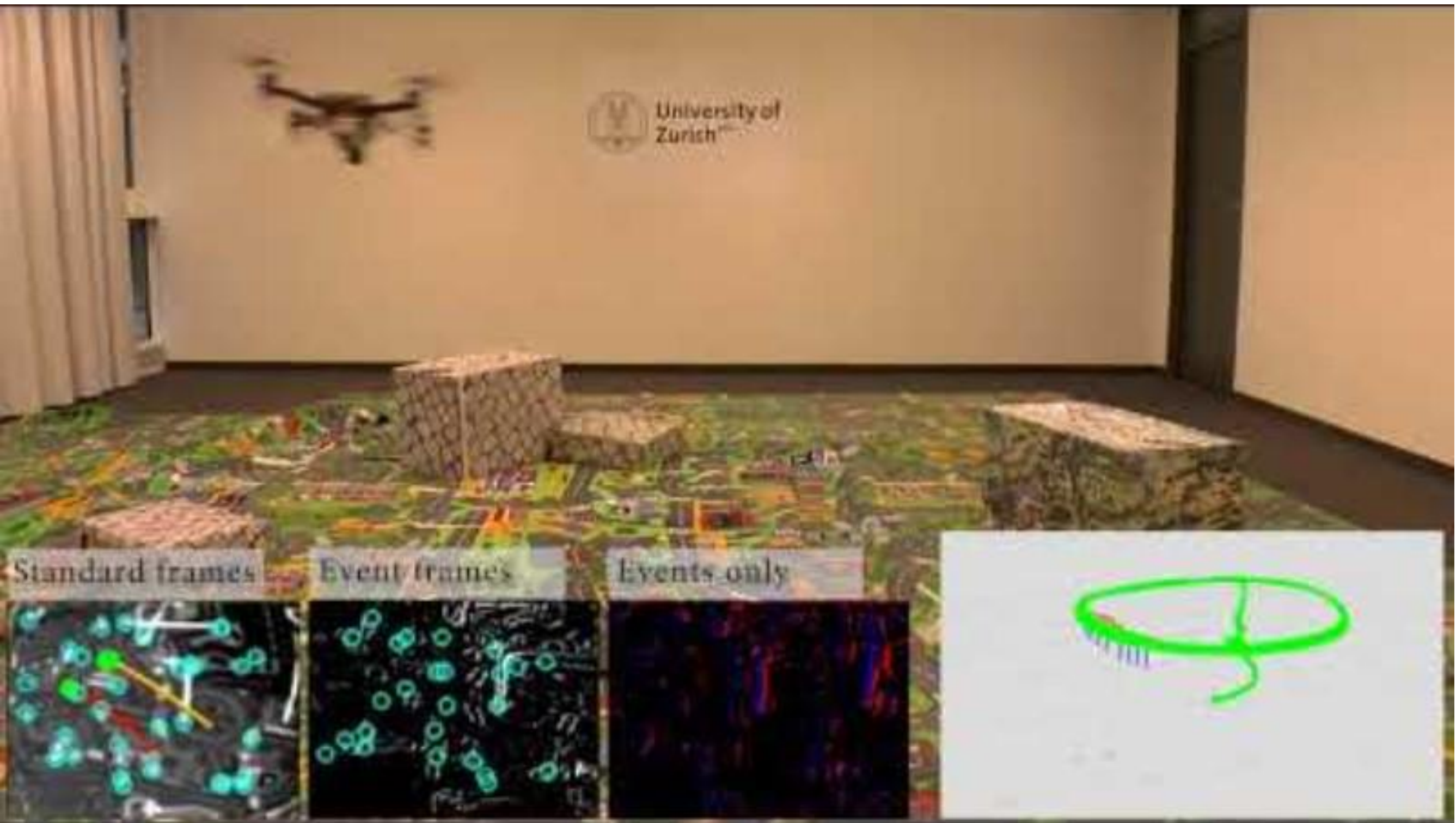


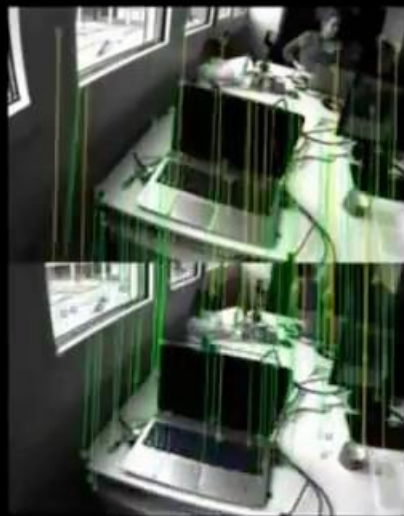
Event-based Cameras for SLAM



Antoni Rosinol Vidal, Henri Rebecq, Timo Horstschafer, Davide Scaramuzza Ultimate SLAM? Combining Events, Images, and IMU for Robust Visual SLAM in HDR and High Speed Scenarios R-AL 2018.

Event-based Cameras for SLAM





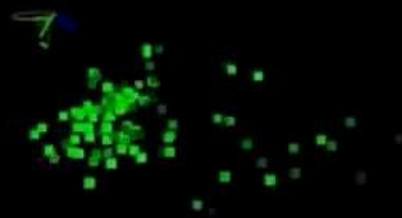
OKVIS



ROVIO



VINS-Mono



Antoni Rosinol Vidal, Henri Rebecq, Timo Horstschaefer, Davide Scaramuzza Ultimate SLAM? Combining Events, Images, and IMU for Robust Visual SLAM in HDR and High Speed Scenarios R-AL 2018.